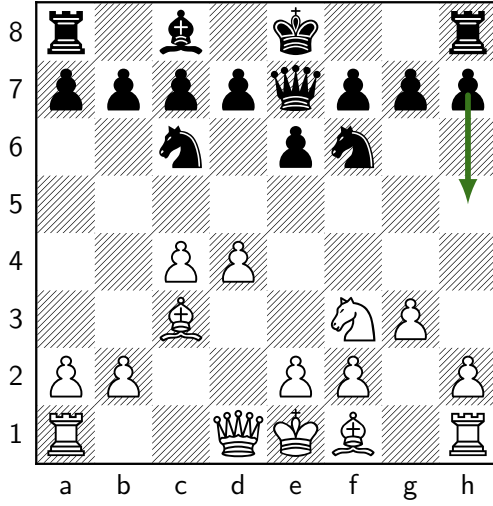


# Melanie-XRAY — Bogo-Indian

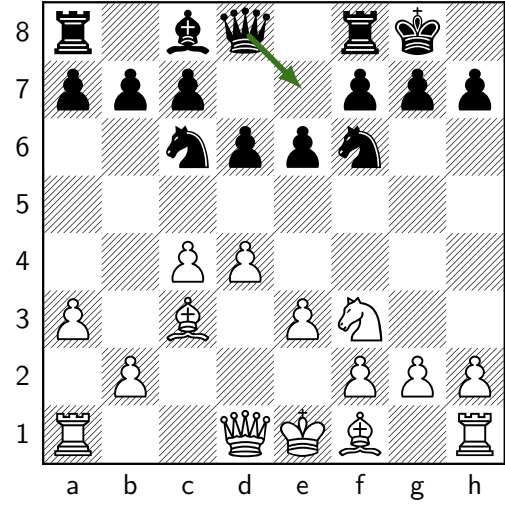
Melanie (Deep\_Vision)

## Abstract

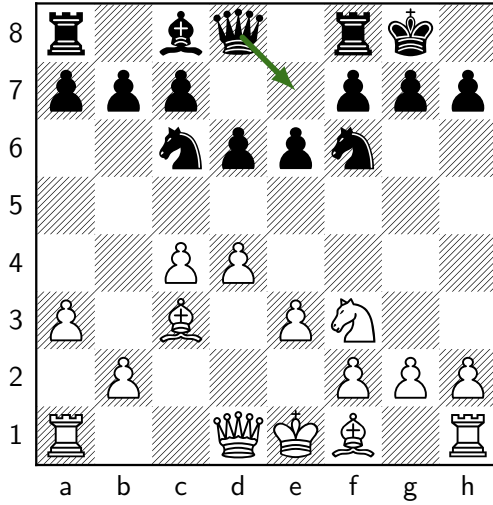
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under  $\mu$  – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



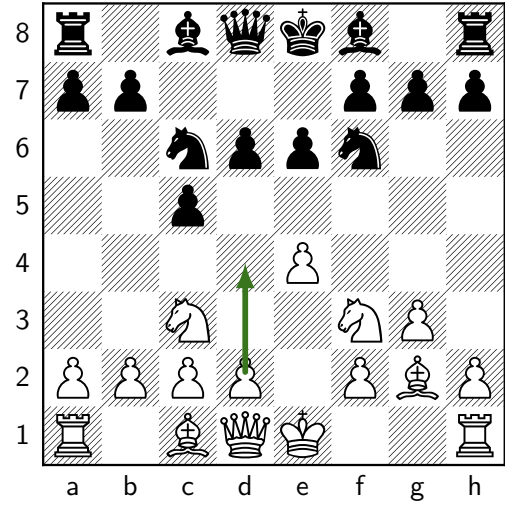
(a)



(b)

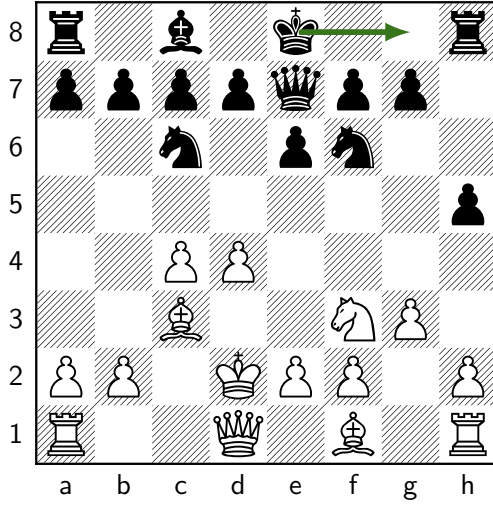


(c)

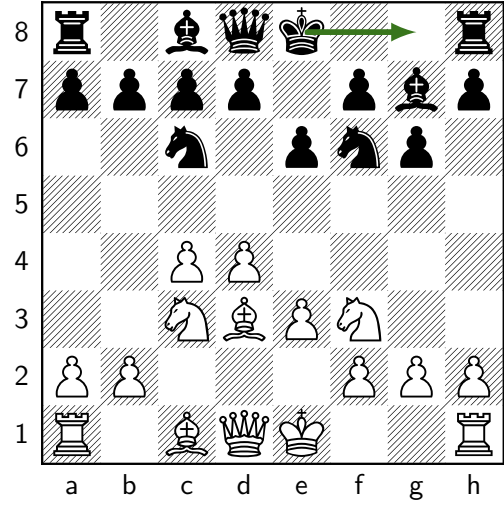


(d)

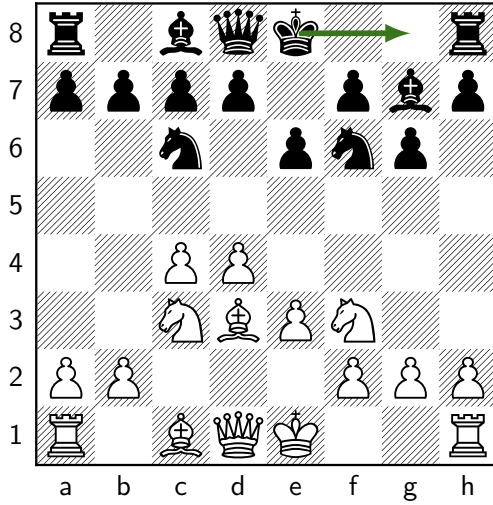
Figure 1: **Move 7, Black: h5**. (a) Our position; Melanie plays h5 (h7h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3433` (ply 15) [structural distance  $d=37$ , QAP obj=31]: the master played Qe7; under  $\mu$  our *Ph2* plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Bc8, our Qe7 as their Rf8; transferred move h7h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3434` (ply 15) [structural distance  $d=37$ , QAP obj=31]: the master played Qe7; under  $\mu$  our *Ph2* plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Bc8, our Qe7 as their Rf8; transferred move h7h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2061` (ply 10) [structural distance  $d=57$ , QAP obj=24]: the master played d4; under  $\mu$  our *Ph2* plays the role of that game’s Pd2; cross-type roles: our Rh8 as their Qd1, our Bc8 as their Pa2; transferred move h7h5 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to h5 (summed QAP objective 304, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



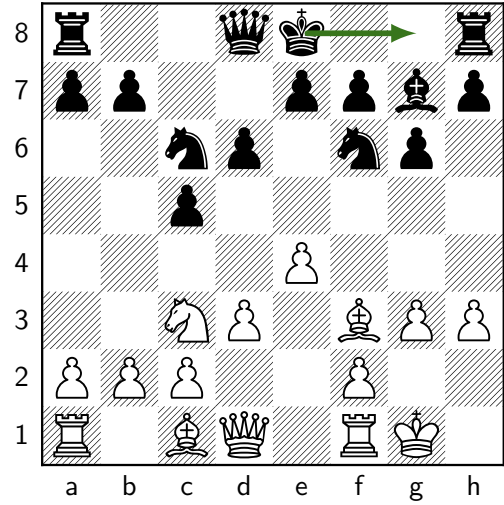
(a)



(b)

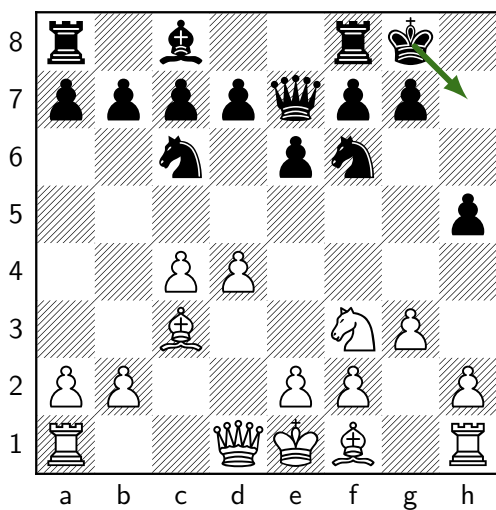


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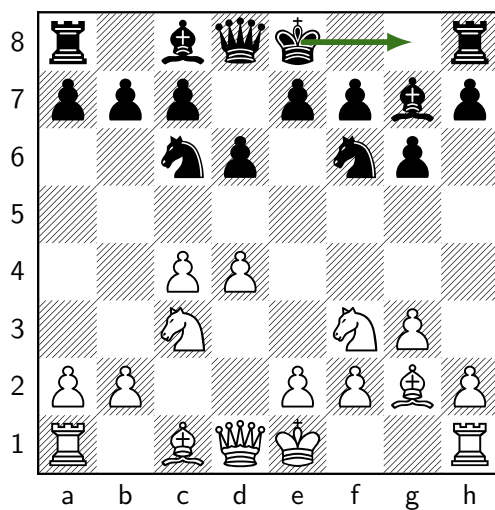


(d)

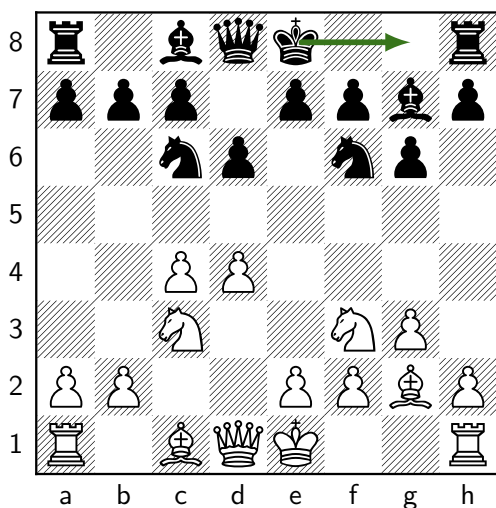
Figure 2: **Move 8, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1925` (ply 11) [structural distance  $d=59$ , QAP obj=30]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pb2* as their *Bc8*, our *Nf6* as their *Rh8*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1926` (ply 11) [structural distance  $d=59$ , QAP obj=30]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *pb2* as their *Bc8*, our *Nf6* as their *Rh8*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2823` (ply 17) [structural distance  $d=57$ , QAP obj=29]: the master played 0-0; under  $\mu$  our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Nf6* as their *Pa7*, our *nf3* as their *Bg7*; transferred move e8g8 (**endorses this move**). Synthesis: 17 retrieved analogs transferred to 0-0 (summed QAP objective 456, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



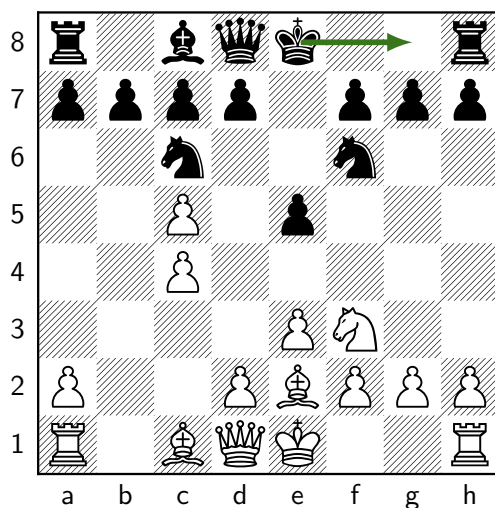
(a)



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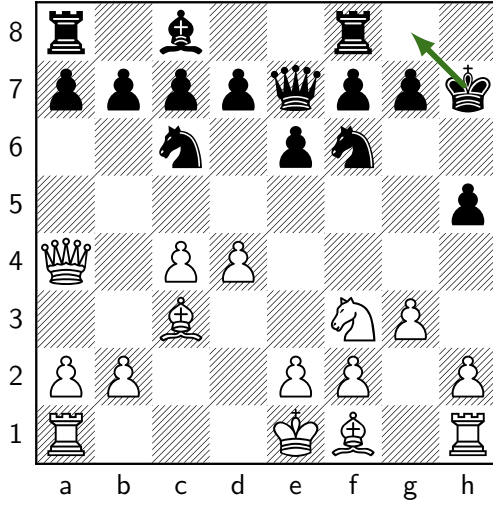


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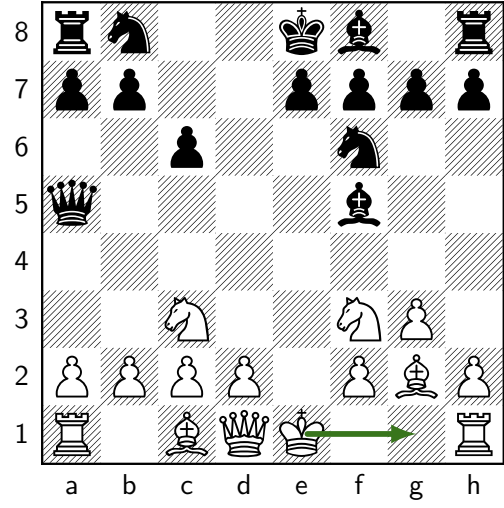


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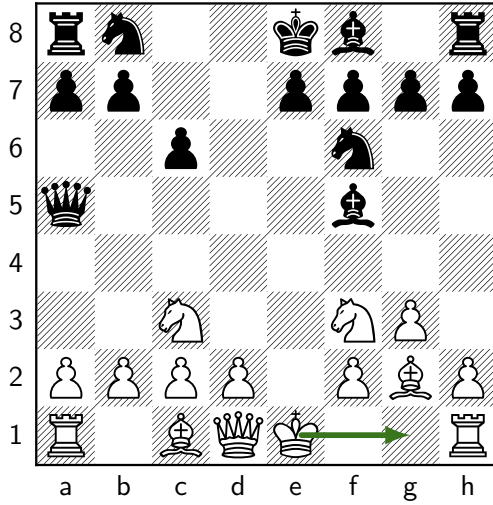
Figure 3: **Move 9, Black: Kh7.** (a) Our position; Melanie plays Kh7 (g8h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#362` (ply 11) [structural distance  $d=59$ , QAP obj=27]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *nf3* as their *Bc8*; transferred move g8h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#363` (ply 11) [structural distance  $d=59$ , QAP obj=27]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *nf3* as their *Bc8*; transferred move g8h7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#666` (ply 15) [structural distance  $d=50$ , QAP obj=26]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pe6* as their *Ra8*, our *pb2* as their *Bc8*; transferred move g8h7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kh7 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



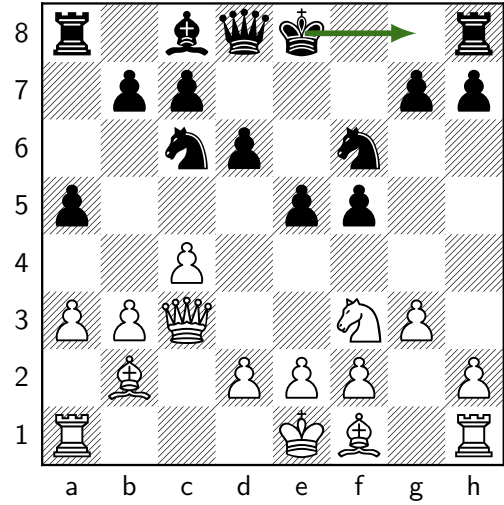
(a)



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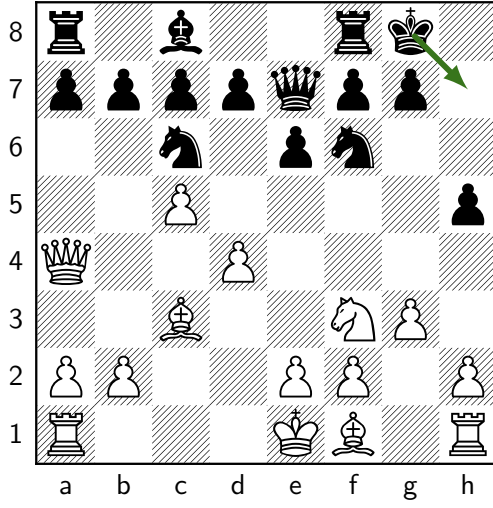


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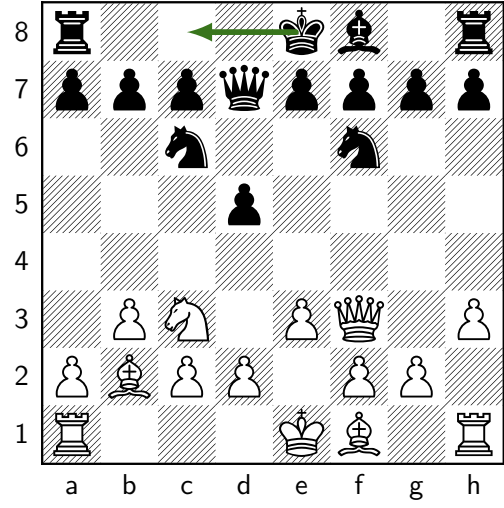


(d)

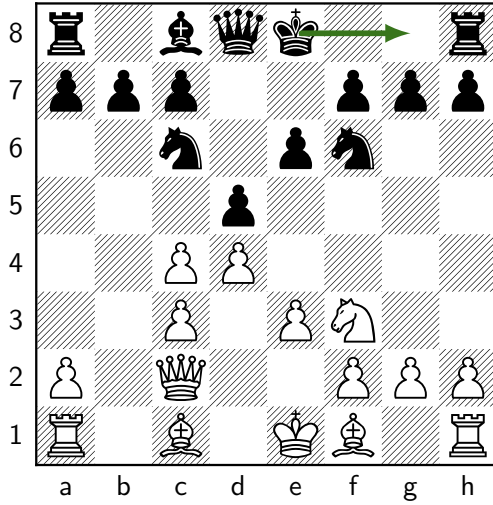
Figure 4: **Move 10, Black: Kg8**. (a) Our position; Melanie plays Kg8 (h7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#702` (ply 12) [structural distance  $d=83$ , QAP obj=93]: the master played 0-0; under  $\mu$  our *Kh2* plays the role of that game’s *Ke1*; cross-type roles: our *Ph5* as their *Bc1*, our *bc3* as their *Pc2*; transferred move h7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#703` (ply 12) [structural distance  $d=83$ , QAP obj=93]: the master played 0-0; under  $\mu$  our *Kh2* plays the role of that game’s *Ke1*; cross-type roles: our *Ph5* as their *Bc1*, our *bc3* as their *Pc2*; transferred move h7g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#21` (ply 17) [structural distance  $d=77$ , QAP obj=39]: the master played 0-0; under  $\mu$  our *Kh2* plays the role of that game’s *Ke1*; cross-type roles: our *pg3* as their *Ra8*, our *Pg7* as their *Qd8*; transferred move h7g8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg8 (summed QAP objective 225, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



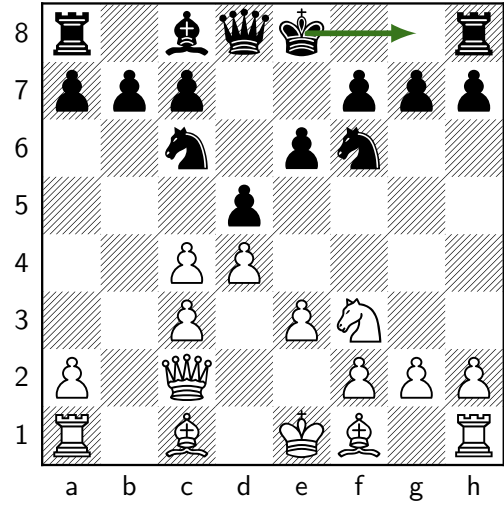
(a)



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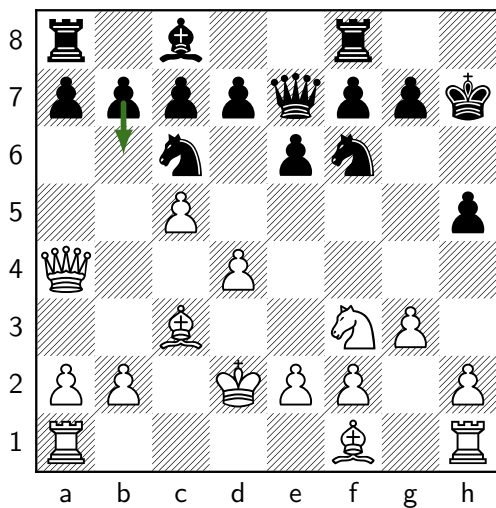


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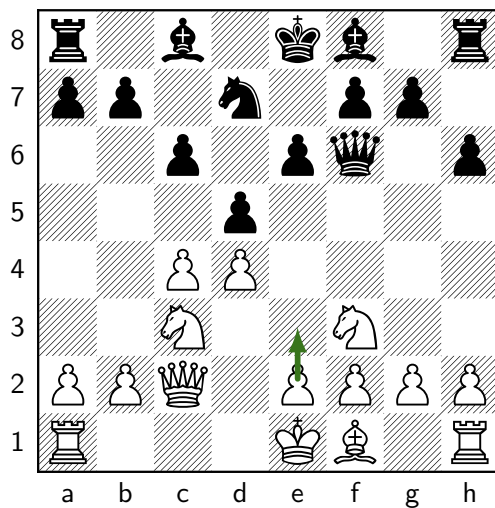


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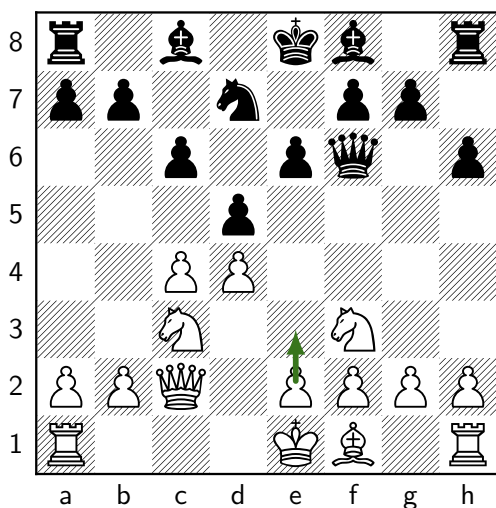
Figure 5: **Move 11, Black: Kh7.** (a) Our position; Melanie plays Kh7 (g8h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3051` (ply 13) [structural distance  $d=79$ , QAP obj=63]: the master played 0-0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Pf7* as their *Bf8*; transferred move g8h7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#461` (ply 13) [structural distance  $d=65$ , QAP obj=28]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pf2* as their *Ra8*, our *ph2* as their *Bc8*; transferred move g8h7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#462` (ply 13) [structural distance  $d=65$ , QAP obj=28]: the master played 0-0; under  $\mu$  our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *pf2* as their *Ra8*, our *ph2* as their *Bc8*; transferred move g8h7 (**endorses this move**). Synthesis: 20 retrieved analogs transferred to Kh7 (summed QAP objective 428, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



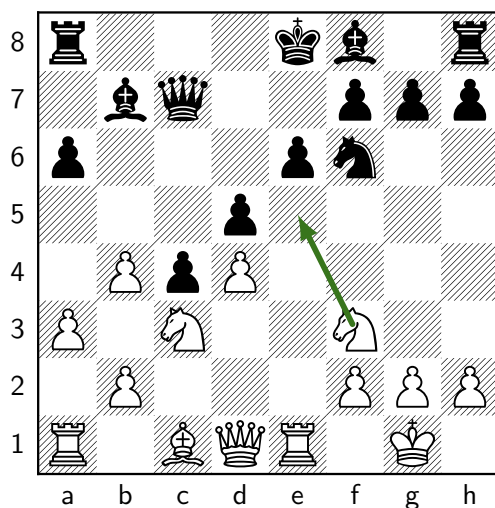
(a)



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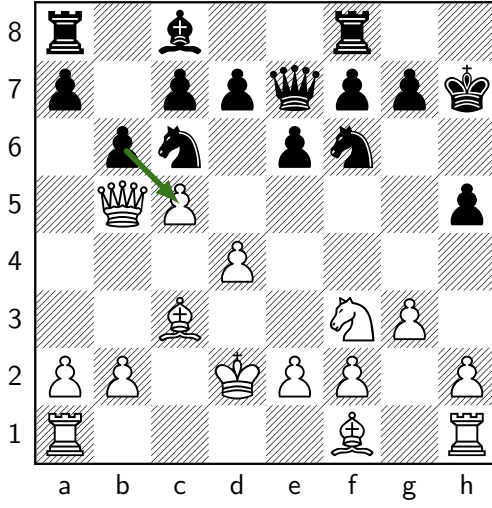


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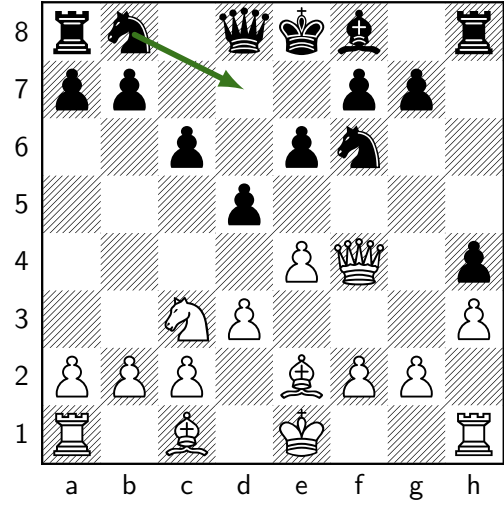


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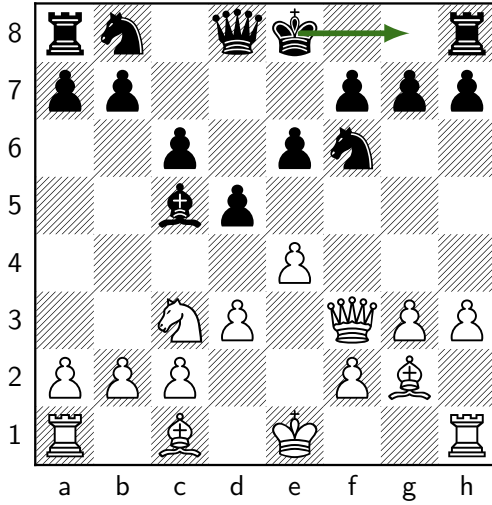
Figure 6: **Move 12, Black: b6.** (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1105` (ply 14) [structural distance  $d=77$ , QAP obj=66]: the master played e3; under  $\mu$  our  $Pb2$  plays the role of that game’s  $Pe2$ ; cross-type roles: our  $Pe6$  as their  $Bf1$ , our  $Bc8$  as their  $Rh1$ ; transferred move b7b6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1106` (ply 14) [structural distance  $d=77$ , QAP obj=66]: the master played e3; under  $\mu$  our  $Pb2$  plays the role of that game’s  $Pe2$ ; cross-type roles: our  $Pe6$  as their  $Bf1$ , our  $Bc8$  as their  $Rh1$ ; transferred move b7b6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#207` (ply 32) [structural distance  $d=71$ , QAP obj=26]: the master played Ne5; under  $\mu$  our  $Pb2$  plays the role of that game’s  $Nf3$  (a **cross-type** correspondence); cross-type roles: our  $Nc6$  as their  $Ra1$ , our  $Pf7$  as their  $Bc1$ ; transferred move b7b6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to b6 (summed QAP objective 159, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



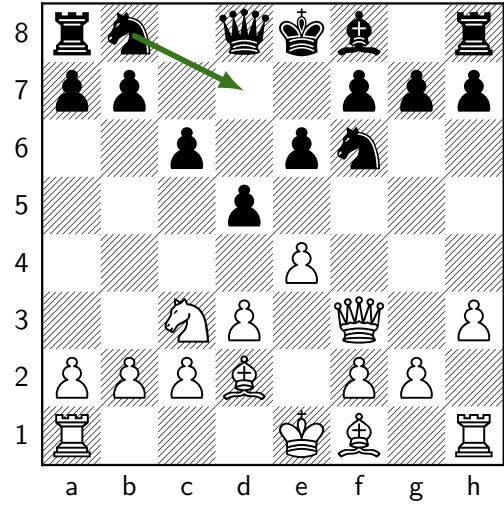
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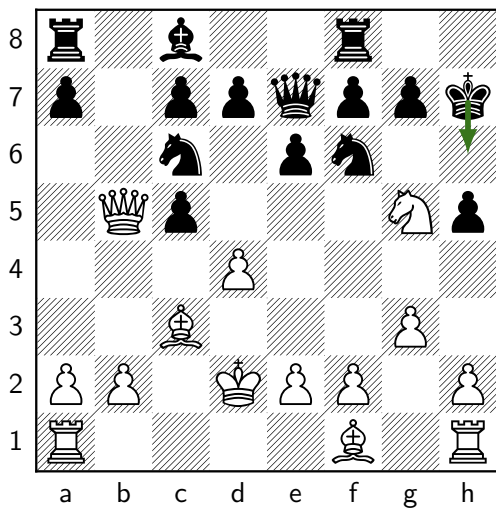
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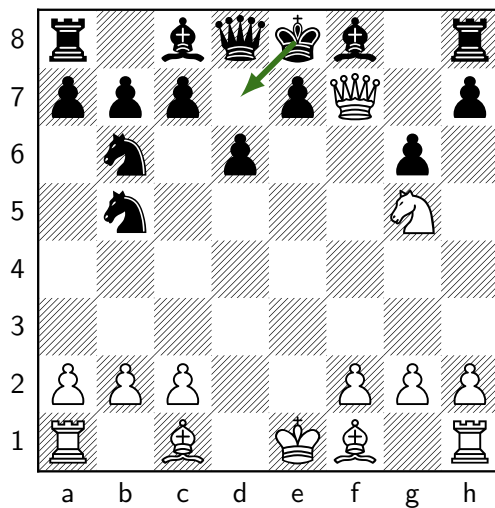
(d)

Figure 7: **Move 13, Black: bxc5**. (a) Our position; Melanie plays **bxc5** (**b6c5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2148` (ply 17) [structural distance  $d=78$ , QAP obj=57]: the master played **Nbd7**; under  $\mu$  our **Pb3** plays the role of that game’s **Nb1** (a **cross-type** correspondence); cross-type roles: our **Nf6** as their **Qd8**, our **pg3** as their **Bf8**; transferred move **b6c5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1478` (ply 15) [structural distance  $d=82$ , QAP obj=50]: the master played **0-0**; under  $\mu$  our **Kh2** plays the role of that game’s **Ke1**; cross-type roles: our **pa2** as their **Nb8**, our **Pg7** as their **Qd8**; transferred move **h7g8**. (d) Retrieved from `ccr1_top_p0002#2347` (ply 13) [structural distance  $d=65$ , QAP obj=50]: the master played **Nbd7**; under  $\mu$  our **Ph4** plays the role of that game’s **Nb1** (a **cross-type** correspondence); cross-type roles: our **Qe7** as their **Ra8**, our **Nf6** as their **Qd8**; transferred move **h5h4**. Synthesis: 1 retrieved analog transferred to **bxc5** (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

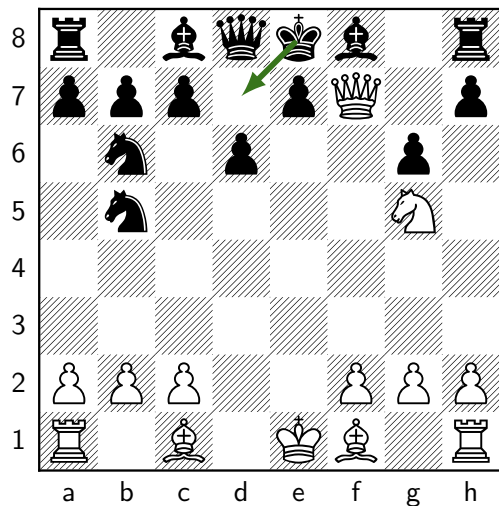




(a)

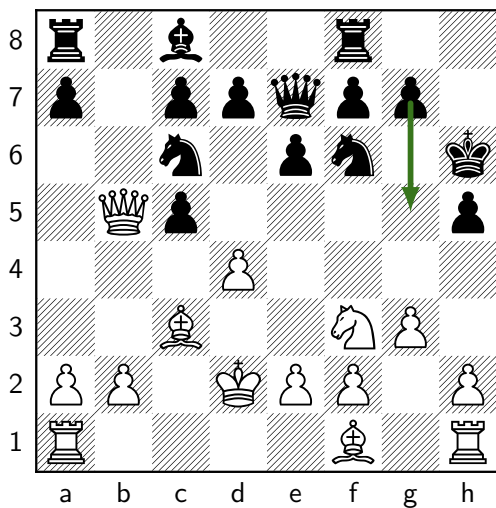


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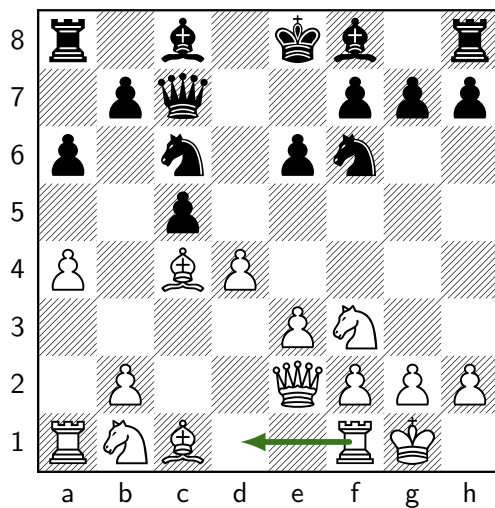


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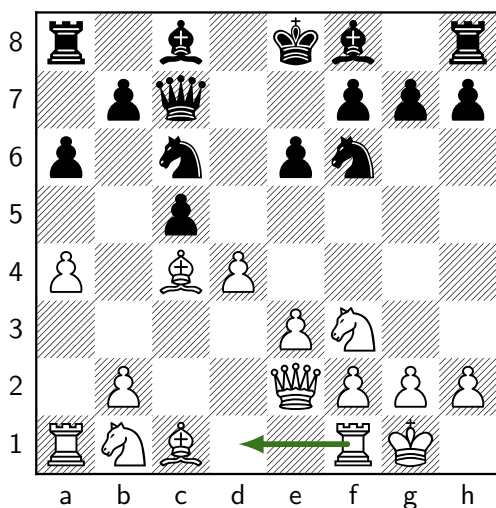
Figure 8: **Move 14, Black: Kh6.** (a) Our position; Melanie plays Kh6 (**h7h6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1329` (ply 23) [structural distance  $d=107$ , QAP obj=38]: the master played Kd7; under  $\mu$  our Kh2 plays the role of that game’s Ke1; cross-type roles: our Pf7 as their Ra8, our Pd7 as their Bc8; transferred move **h7g8**. (c) Retrieved from `ccr1_top_p0002#2827` (ply 23) [structural distance  $d=107$ , QAP obj=38]: the master played Kd7; under  $\mu$  our Kh2 plays the role of that game’s Ke1; cross-type roles: our Pf7 as their Ra8, our Pd7 as their Bc8; transferred move **h7g8**. Synthesis: no retrieved analog transferred to Kh6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



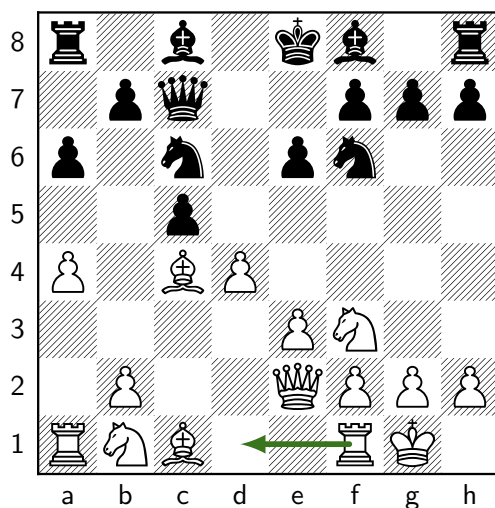
(a)



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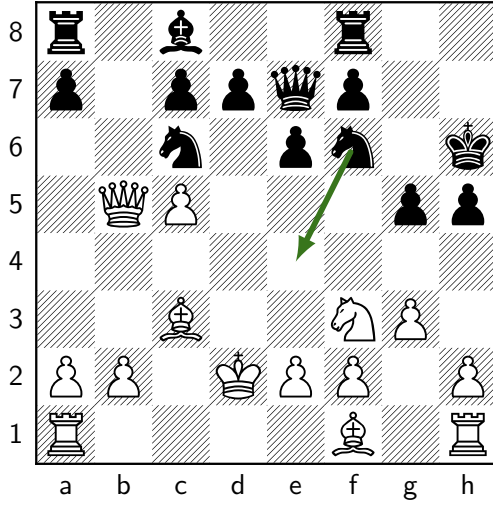


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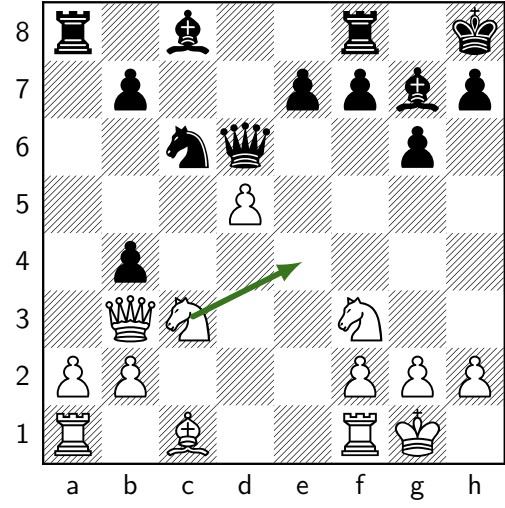


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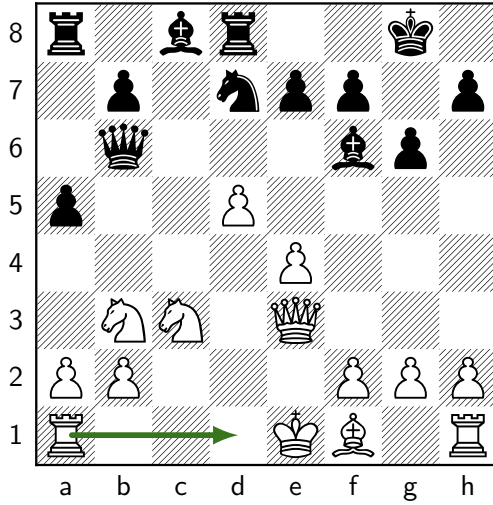
Figure 9: **Move 15, Black: g5**. (a) Our position; Melanie plays g5 (g7g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1039` (ply 16) [structural distance  $d=82$ , QAP obj=37]: the master played Rd1; under  $\mu$  our Pg2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Nb1, our Nf6 as their Bc1; transferred move g7g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3084` (ply 16) [structural distance  $d=82$ , QAP obj=37]: the master played Rd1; under  $\mu$  our Pg2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Nb1, our Nf6 as their Bc1; transferred move g7g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3912` (ply 16) [structural distance  $d=82$ , QAP obj=37]: the master played Rd1; under  $\mu$  our Pg2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Pc7 as their Nb1, our Nf6 as their Bc1; transferred move g7g5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to g5 (summed QAP objective 148, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



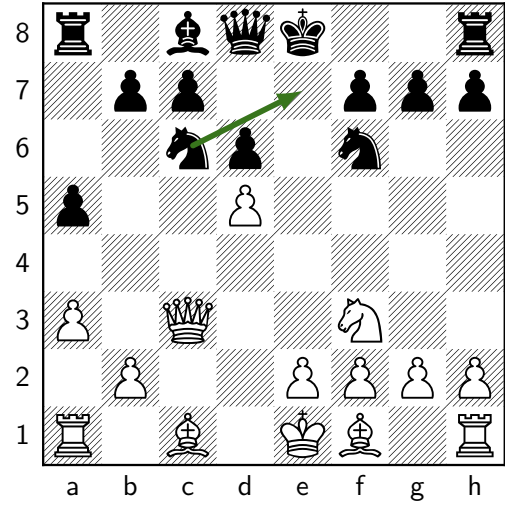
(a)



(b)

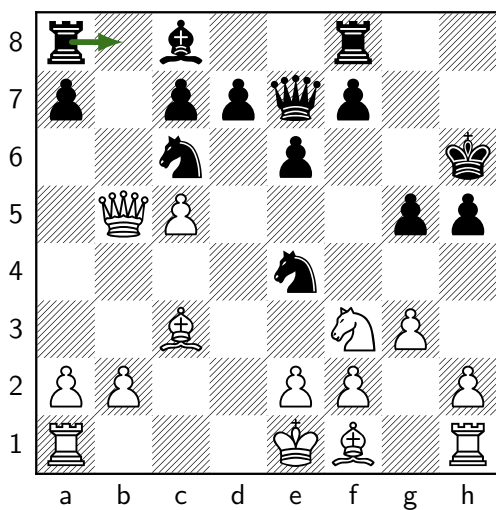


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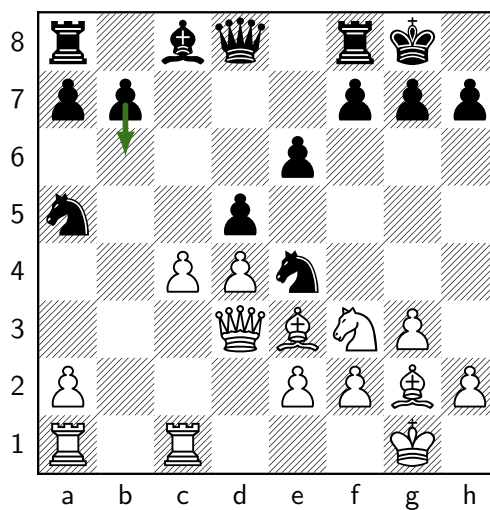


(d)

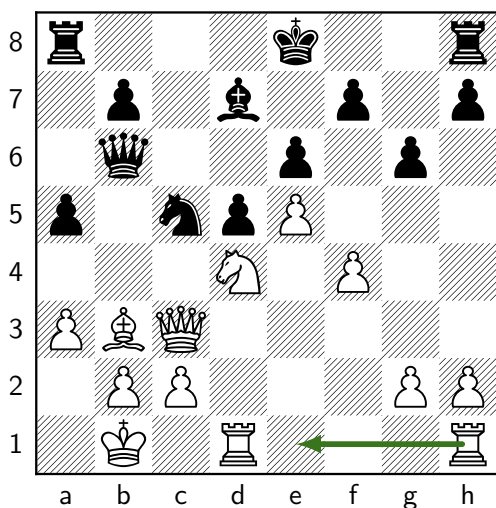
Figure 10: **Move 16, Black: Ne4+.** (a) Our position; Melanie plays Ne4+ (f6e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1296` (ply 28) [structural distance  $d=74$ , QAP obj=32]: the master played Ne4; under  $\mu$  our Nf3 plays the role of that game’s Nc3; cross-type roles: our Pg5 as their Bc1, our bc3 as their pb4; transferred move f6e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3105` (ply 24) [structural distance  $d=81$ , QAP obj=33]: the master played Rd1; under  $\mu$  our Ra1 plays the role of that game’s Ra1; cross-type roles: our ph2 as their Bf1, our Qe7 as their Rh1; transferred move a8b8. (d) Retrieved from `ccr1_top_p0000#857` (ply 17) [structural distance  $d=80$ , QAP obj=33]: the master played Ne7; under  $\mu$  our Nc3 plays the role of that game’s Nc3; cross-type roles: our Qe7 as their Ra8, our nf3 as their Bc8; transferred move c6b8. Synthesis: 1 retrieved analog transferred to Ne4+ (summed QAP objective 32, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



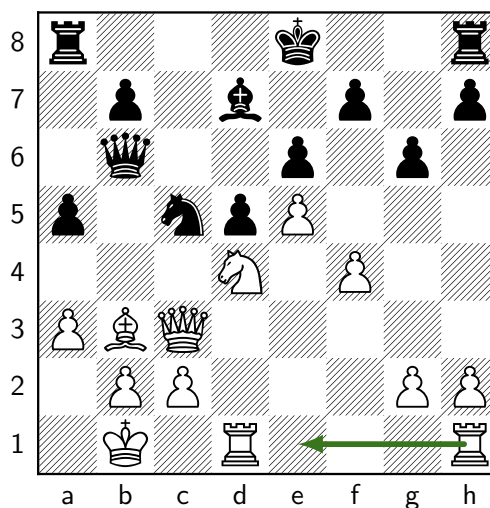
(a)



(b)

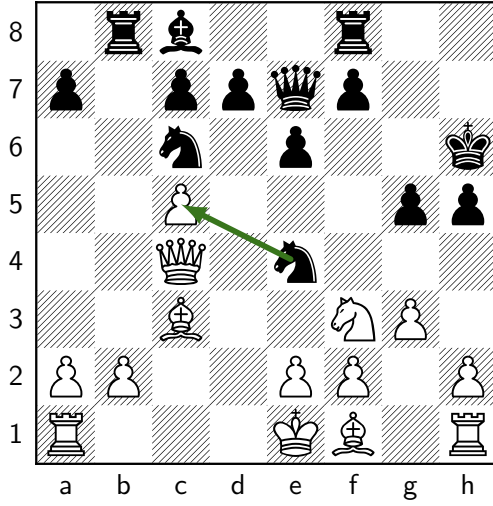


(c)

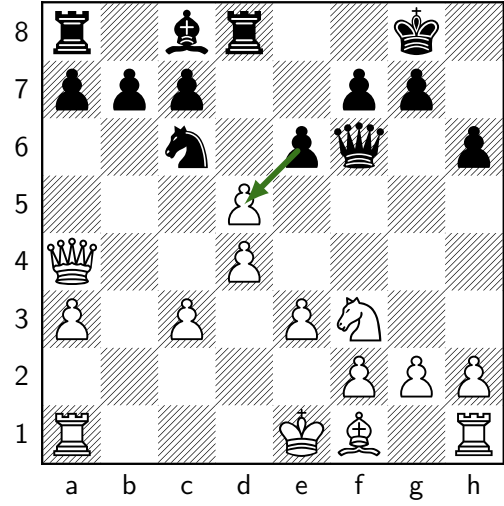


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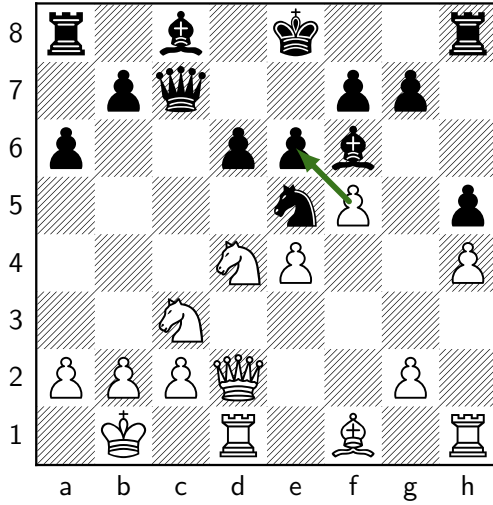
Figure 11: **Move 17, Black: Rb8**. (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1163` (ply 25) [structural distance  $d=94$ , QAP obj=35]: the master played b6; under  $\mu$  our *Ra1* plays the role of that game’s *Pb2* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *Pc7* as their *Bc8*; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#381` (ply 34) [structural distance  $d=100$ , QAP obj=27]: the master played Rhe1; under  $\mu$  our *Ra1* plays the role of that game’s *Rh1*; cross-type roles: our *Qe7* as their *Rd1*, our *Nc6* as their *Bb3*; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2248` (ply 34) [structural distance  $d=100$ , QAP obj=27]: the master played Rhe1; under  $\mu$  our *Ra1* plays the role of that game’s *Rh1*; cross-type roles: our *Qe7* as their *Rd1*, our *Nc6* as their *Bb3*; transferred move a8b8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Rb8 (summed QAP objective 125, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



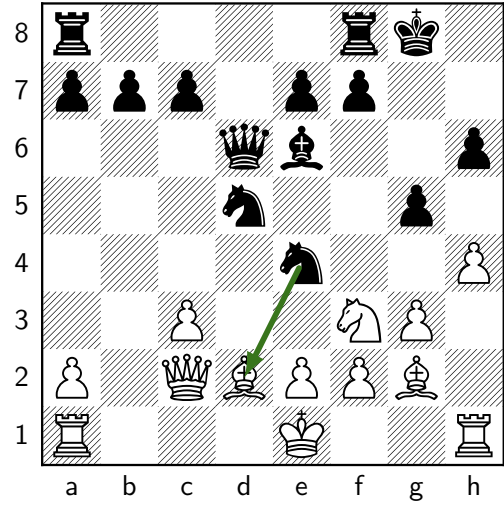
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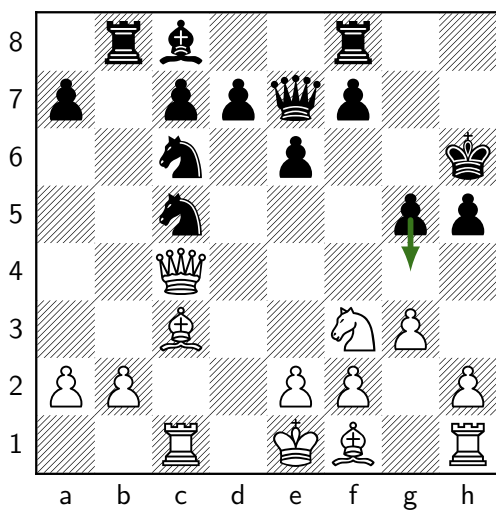


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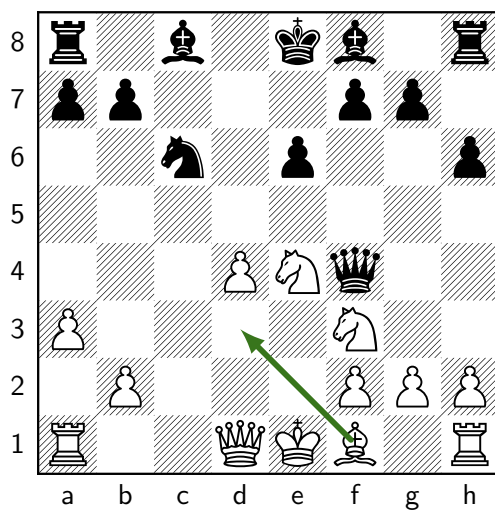


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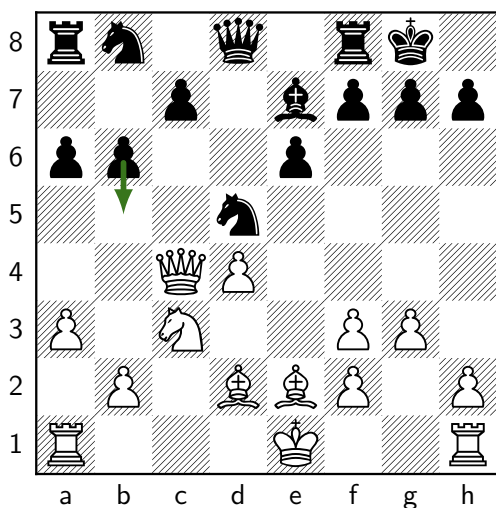
Figure 12: **Move 18, Black: Nxc5.** (a) Our position; Melanie plays Nxc5 (e4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#236` (ply 21) [structural distance  $d=103$ , QAP obj=52]: the master played `exd5`; under  $\mu$  our *Ne5* plays the role of that game’s *Pe3* (a **cross-type** correspondence); cross-type roles: our *bf1* as their *Ra8*, our *Rb8* as their *Bc8*; transferred move `e4c5` (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1828` (ply 30) [structural distance  $d=109$ , QAP obj=38]: the master played `fxe6`; under  $\mu$  our *Ne5* plays the role of that game’s *Pf5* (a **cross-type** correspondence); cross-type roles: our *Pa7* as their *Rd1*, our *Rb8* as their *Bf1*; transferred move `e4c5` (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#369` (ply 35) [structural distance  $d=88$ , QAP obj=131]: the master played `Nxd2`; under  $\mu$  our *Ne5* plays the role of that game’s *Ne5*; cross-type roles: our *Pa7* as their *Ra8*, our *Rb8* as their *Pa7*; transferred move `e4d2`. Synthesis: 2 retrieved analogs transferred to Nxc5 (summed QAP objective 90, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



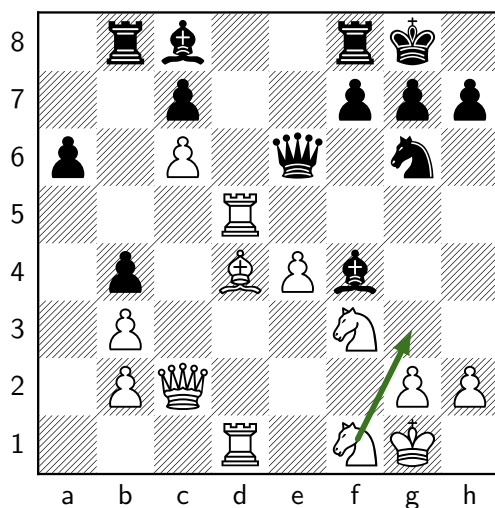
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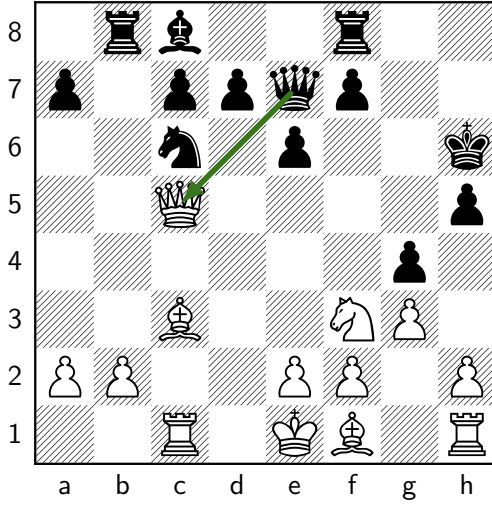


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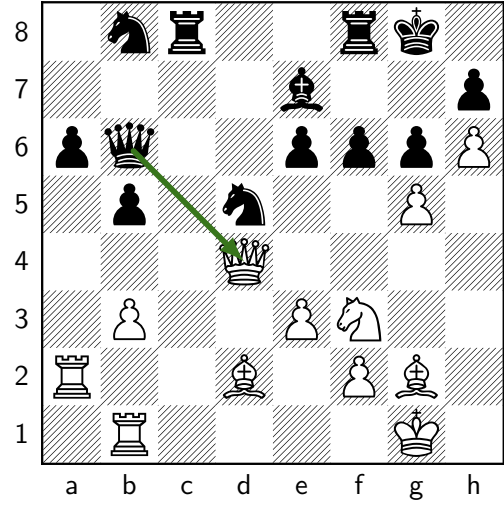


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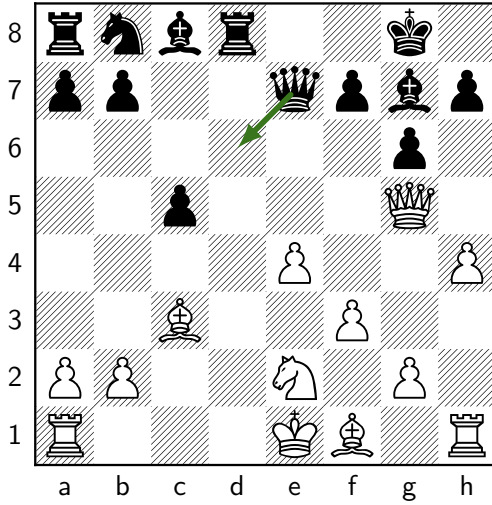
Figure 13: **Move 19, Black: g4**. (a) Our position; Melanie plays g4 (g5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `crr1_top_p0002#48` (ply 20) [structural distance  $d=105$ , QAP obj=52]: the master played Bd3; under  $\mu$  our  $Pg4$  plays the role of that game’s  $Bf1$  (a **cross-type** correspondence); cross-type roles: our  $Pd7$  as their  $Rh1$ , our  $bf1$  as their  $Pb2$ ; transferred move g5g4 (**endorses this move**). (c) Retrieved from `crr1_top_p0000#1862` (ply 25) [structural distance  $d=95$ , QAP obj=35]: the master played b5; under  $\mu$  our  $Pg4$  plays the role of that game’s  $Pb3$ ; cross-type roles: our  $Nc6$  as their  $Ra8$ , our  $Ph5$  as their  $Nb8$ ; transferred move g5g4 (**endorses this move**). (d) Retrieved from `crr1_top_p0002#2252` (ply 46) [structural distance  $d=104$ , QAP obj=20]: the master played Ng3; under  $\mu$  our  $Pg4$  plays the role of that game’s  $Nf1$  (a **cross-type** correspondence); cross-type roles: our  $Bc8$  as their  $Rd1$ , our  $Rb8$  as their  $Pb2$ ; transferred move g5g4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to g4 (summed QAP objective 156, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



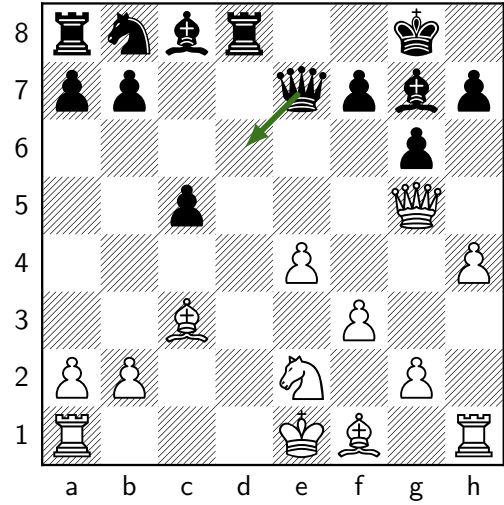
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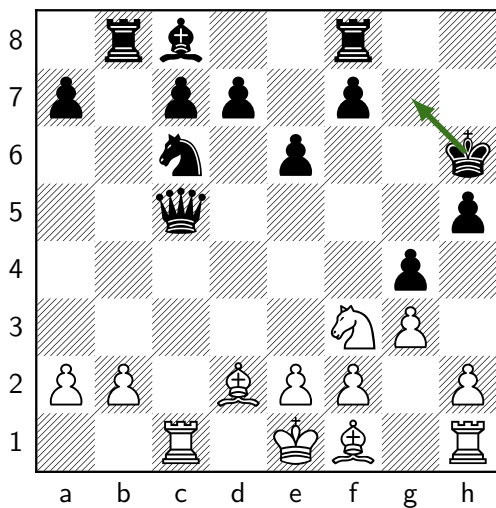
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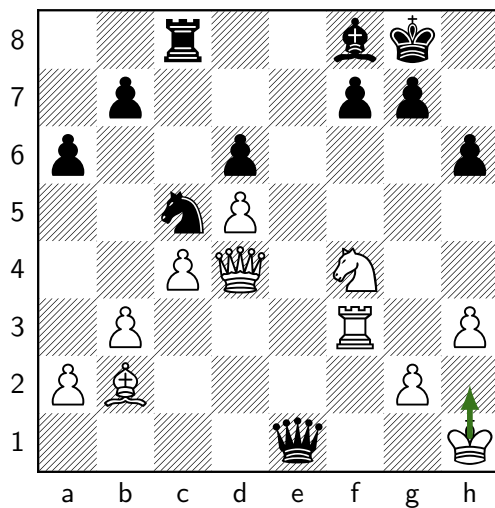
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Figure 14: **Move 20, Black: Qxc5.** (a) Our position; Melanie plays Qxc5 (e7c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0003#659 (ply 53) [structural distance  $d=104$ , QAP obj=67]: the master played Qxd4; under  $\mu$  our Qe2 plays the role of that game’s Qb3; cross-type roles: our Pd7 as their Nb8, our Ph5 as their Rc8; transferred move e7c5 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#240 (ply 25) [structural distance  $d=112$ , QAP obj=62]: the master played Qd6; under  $\mu$  our Qe2 plays the role of that game’s Qe2; cross-type roles: our pa2 as their Ra8, our Rb8 as their Nb8; transferred move e7c5 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#241 (ply 25) [structural distance  $d=112$ , QAP obj=62]: the master played Qd6; under  $\mu$  our Qe2 plays the role of that game’s Qe2; cross-type roles: our pa2 as their Ra8, our Rb8 as their Nb8; transferred move e7c5 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Qxc5 (summed QAP objective 412, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

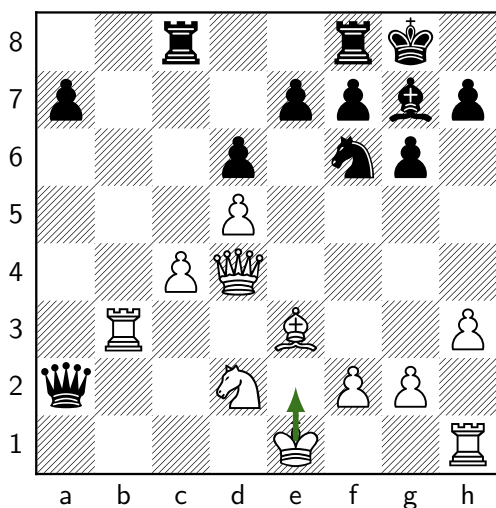




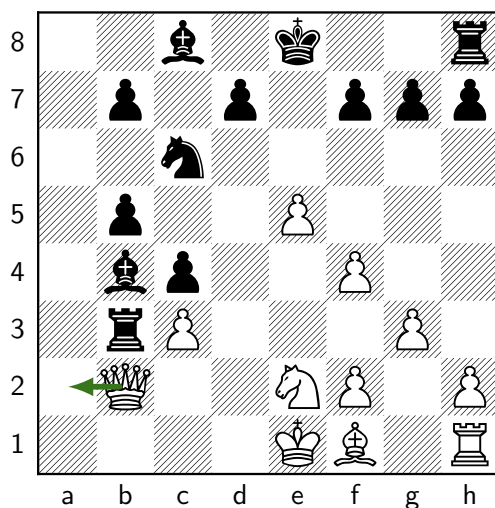
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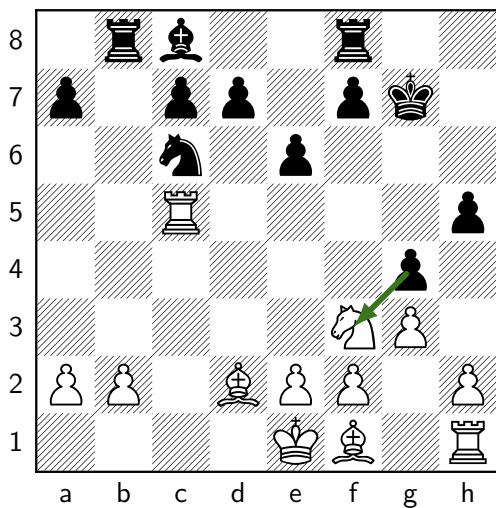
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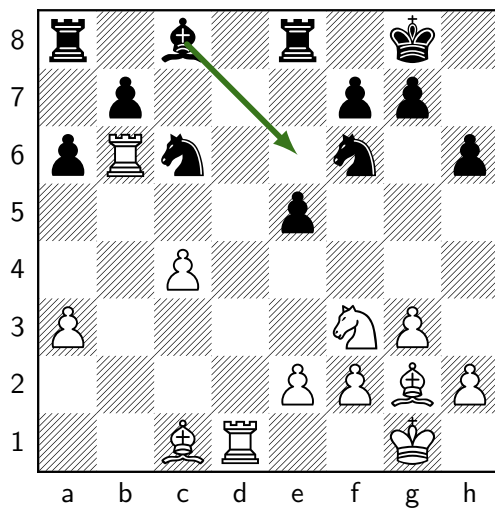
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Figure 15: **Move 21, Black: Kg7.** (a) Our position; Melanie plays Kg7 (**h6g7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3409` (ply 50) [structural distance  $d=119$ , QAP obj=55]: the master played Kh2; under  $\mu$  our Kh3 plays the role of that game’s Kh1; cross-type roles: our bd2 as their qe1, our Qc5 as their Bb2; transferred move h6g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#270` (ply 36) [structural distance  $d=113$ , QAP obj=7]: the master played Ke2; under  $\mu$  our Kh3 plays the role of that game’s Ke1; cross-type roles: our Ph5 as their Rh1, our pa2 as their qa2; transferred move h6g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2993` (ply 32) [structural distance  $d=112$ , QAP obj=6]: the master played Qa2; under  $\mu$  our Kh3 plays the role of that game’s Qb2 (a **cross-type** correspondence); cross-type roles: our Pg4 as their Bf1, our Pe6 as their Rh1; transferred move h6g7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg7 (summed QAP objective 73, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

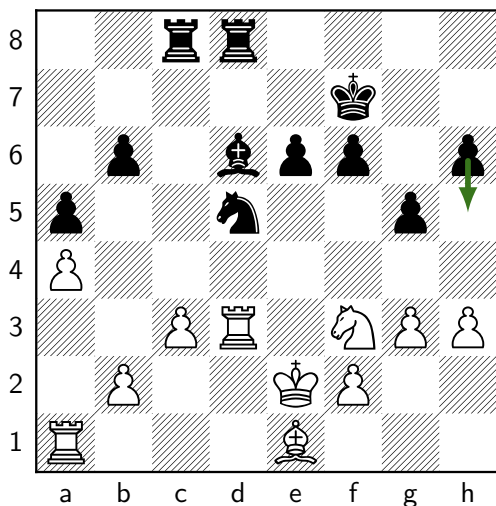




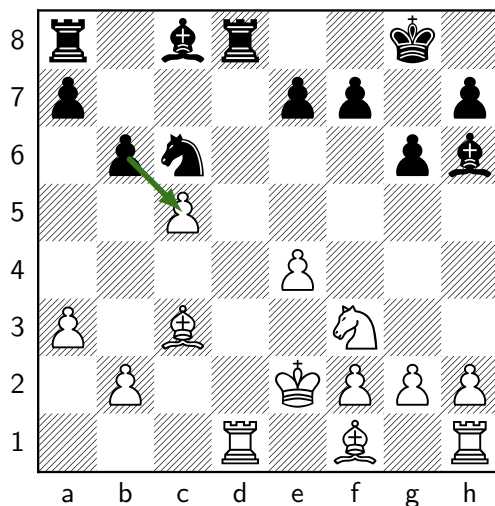
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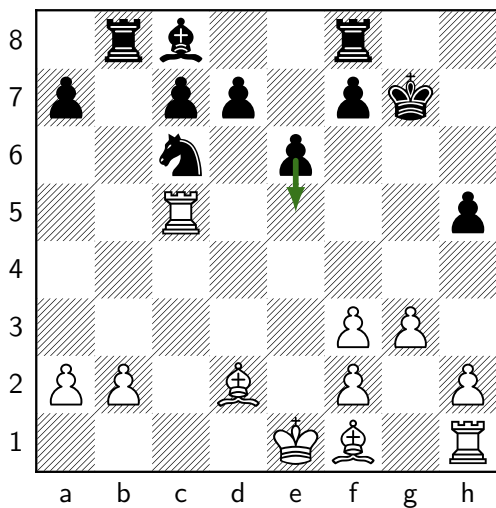


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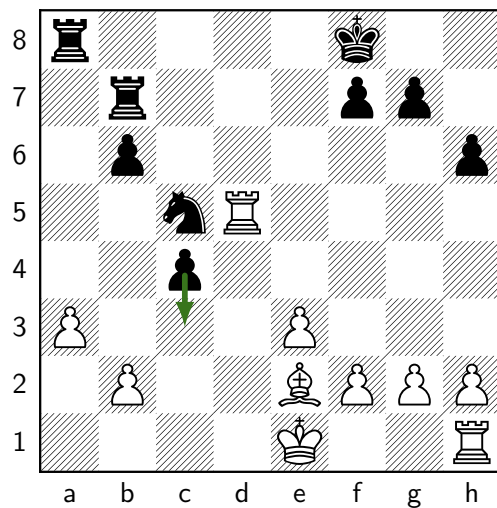


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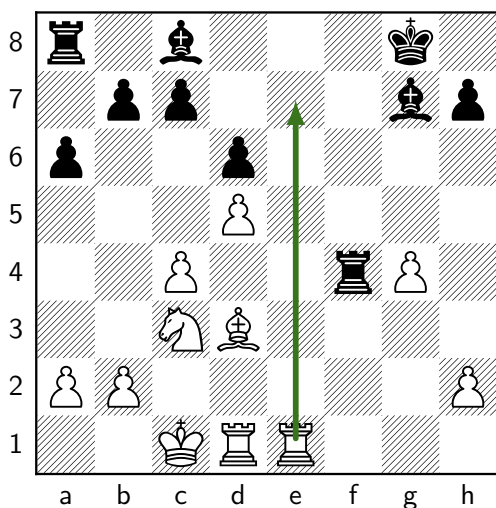
Figure 16: **Move 22, Black: gxf3.** (a) Our position; Melanie plays **gxf3** (**g4f3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1411` (ply 33) [structural distance  $d=70$ , QAP obj=27]: the master played **Be6**; under  $\mu$  our *Pg5* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Pe6* as their *Ra8*, our *Nc6* as their *Pb7*; transferred move **g4f3** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2216` (ply 51) [structural distance  $d=74$ , QAP obj=15]: the master played **h5**; under  $\mu$  our *Pg5* plays the role of that game’s *Ph3*; cross-type roles: our *Bc8* as their *Pb6*, our *Pc7* as their *Bd6*; transferred move **g4f3** (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2273` (ply 27) [structural distance  $d=70$ , QAP obj=14]: the master played **bxc5**; under  $\mu$  our *Pg5* plays the role of that game’s *Pb3*; cross-type roles: our *Pc7* as their *Ra8*, our *Rb8* as their *Bc8*; transferred move **g4f3** (**endorses this move**). Synthesis: 10 retrieved analogs transferred to **gxf3** (summed QAP objective 115, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



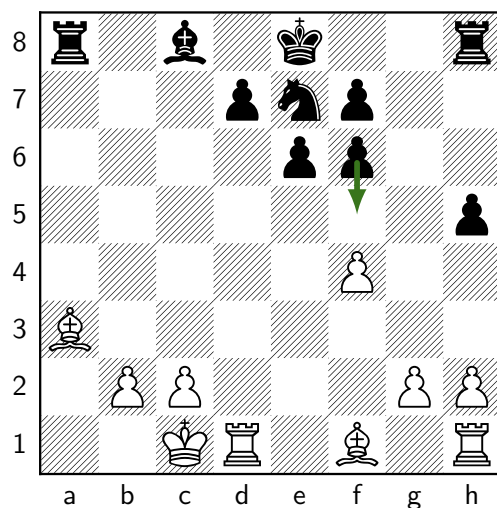
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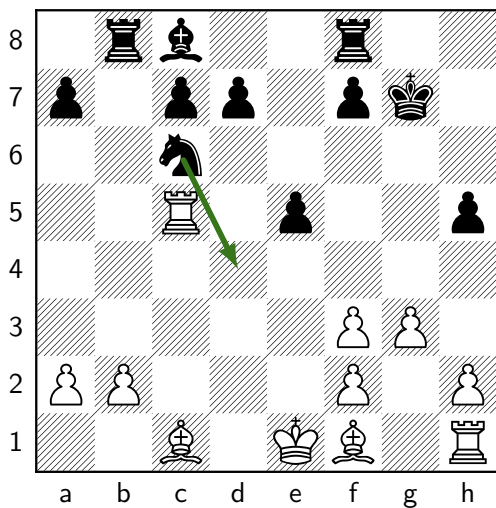


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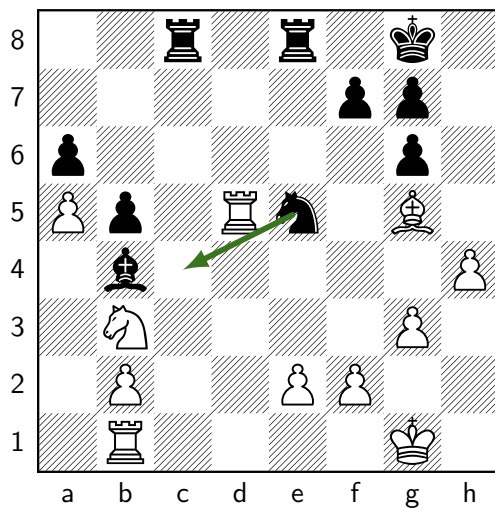


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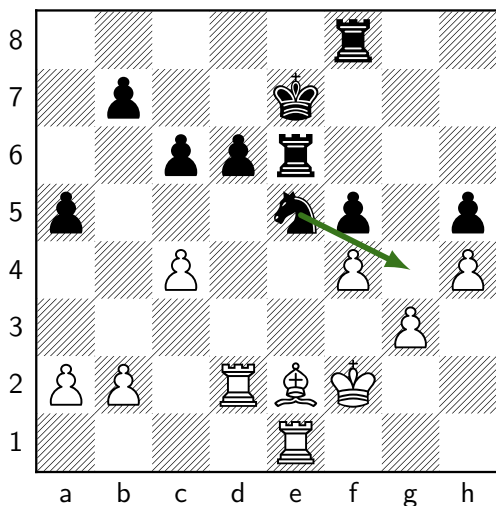
Figure 17: **Move 23, Black: e5**. (a) Our position; Melanie plays **e5** (**e6e5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2050` (ply 49) [structural distance  $d=55$ , QAP obj=23]: the master played **c3**; under  $\mu$  our *Pe3* plays the role of that game’s *Pc5*; cross-type roles: our *Ph5* as their *Ra8*, our *Bc8* as their *Pf7*; transferred move **e6e5** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3790` (ply 42) [structural distance  $d=57$ , QAP obj=10]: the master played **Re7**; under  $\mu$  our *Pe3* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *ph2* as their *Rd1*, our *Rb8* as their *Pa2*; transferred move **e6e5** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3982` (ply 45) [structural distance  $d=52$ , QAP obj=9]: the master played **f5**; under  $\mu$  our *Pe3* plays the role of that game’s *Pf3*; cross-type roles: our *pg3* as their *Ra8*, our *pf2* as their *Rh8*; transferred move **e6e5** (**endorses this move**). Synthesis: 6 retrieved analogs transferred to **e5** (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



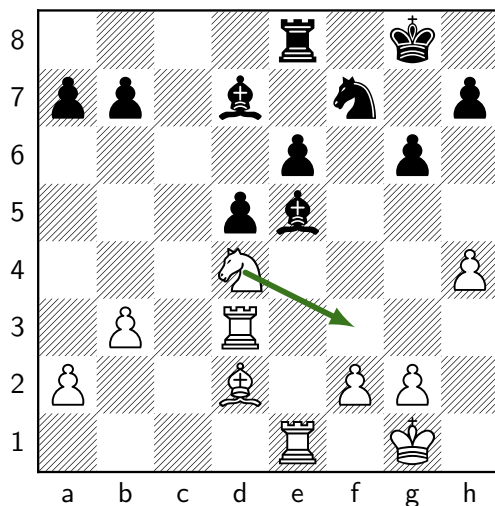
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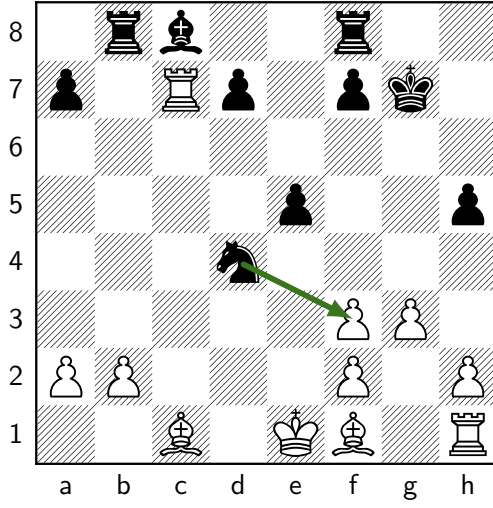


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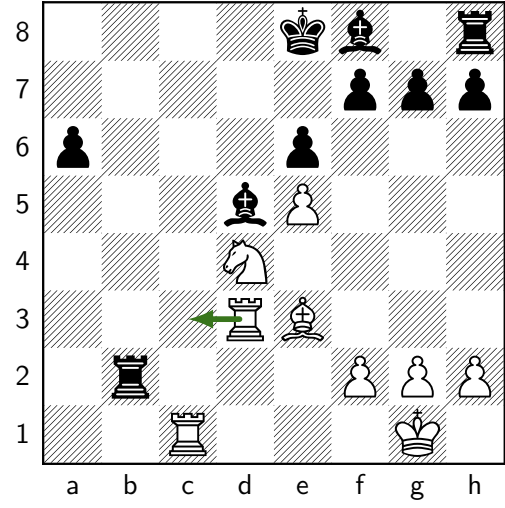


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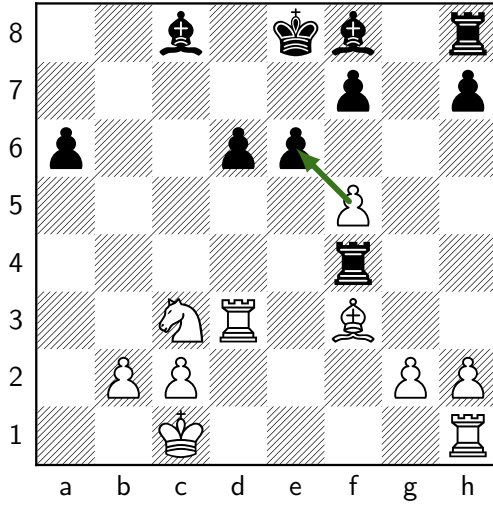
Figure 18: **Move 24, Black: Nd4.** (a) Our position; Melanie plays Nd4 (c6d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0002#830 (ply 61) [structural distance  $d=59$ , QAP obj=25]: the master played Nc4; under  $\mu$  our Nc3 plays the role of that game’s Ne4; cross-type roles: our pf3 as their Rc8, our ph2 as their Re8; transferred move c6d4 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#1547 (ply 63) [structural distance  $d=60$ , QAP obj=24]: the master played Ng4+; under  $\mu$  our Nc3 plays the role of that game’s Ne4; cross-type roles: our Pe5 as their Rf8, our Rb8 as their Pc6; transferred move c6d4 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0000#3625 (ply 56) [structural distance  $d=60$ , QAP obj=23]: the master played Nf3; under  $\mu$  our Nc3 plays the role of that game’s Nd4; cross-type roles: our pf3 as their Re1, our Rf8 as their Pf2; transferred move c6d4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nd4 (summed QAP objective 79, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



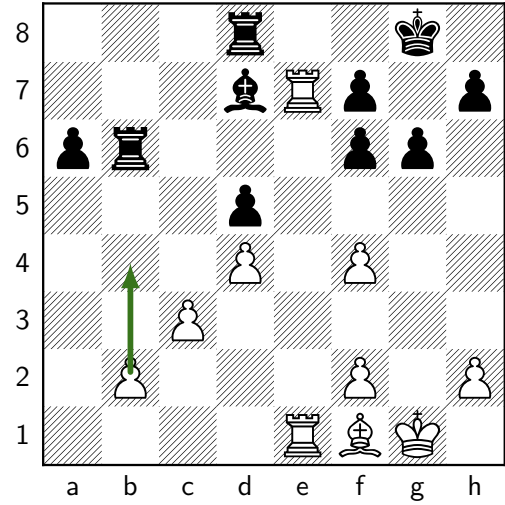
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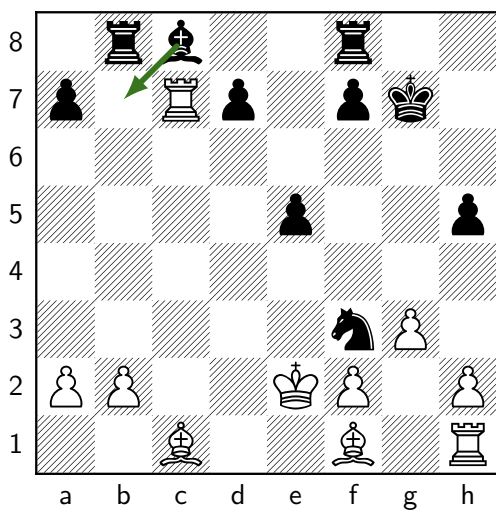


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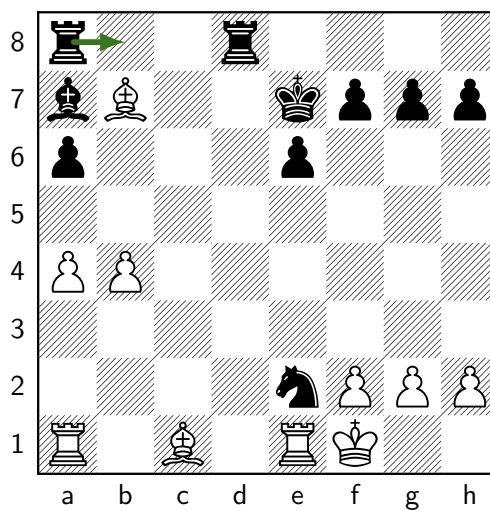


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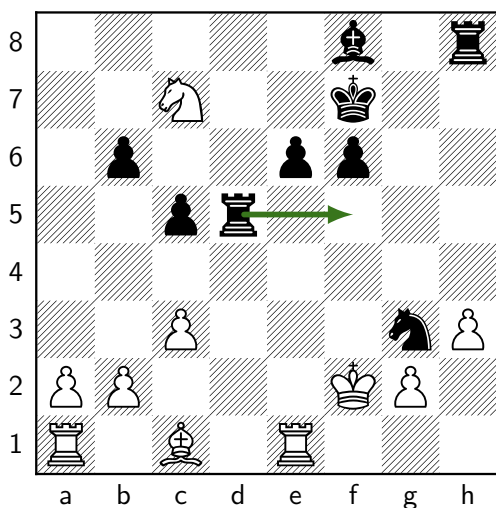
Figure 19: **Move 25, Black: Nx f3+.** (a) Our position; Melanie plays Nx f3+ (d4 f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0002#3782` (ply 46) [structural distance  $d=51$ , QAP obj=9]: the master played R d c3; under  $\mu$  our Nd5 plays the role of that game’s Rd3 (a **cross-type** correspondence); cross-type roles: our Pd7 as their Rc1, our Bc8 as their Pf2; transferred move d4 f3 (**endorses this move**). (c) Retrieved from `ccl1_top_p0003#1634` (ply 34) [structural distance  $d=45$ , QAP obj=38]: the master played f x e6; under  $\mu$  our Pd2 plays the role of that game’s Pf5; cross-type roles: our rc7 as their pe6, our pf2 as their bf8; transferred move d7 d6. (d) Retrieved from `ccl1_top_p0001#284` (ply 46) [structural distance  $d=56$ , QAP obj=36]: the master played b4; under  $\mu$  our Pa2 plays the role of that game’s Pb2; cross-type roles: our bc1 as their Re1, our Nd4 as their Bf1; transferred move a7 a6. Synthesis: 1 retrieved analog transferred to Nx f3+ (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



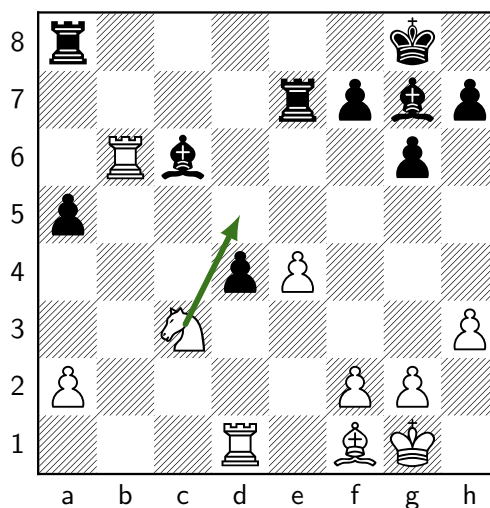
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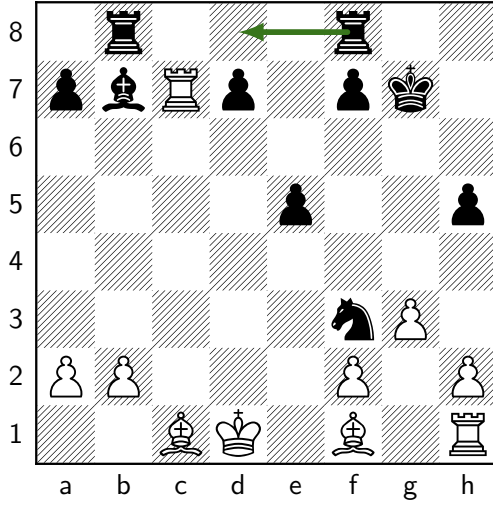


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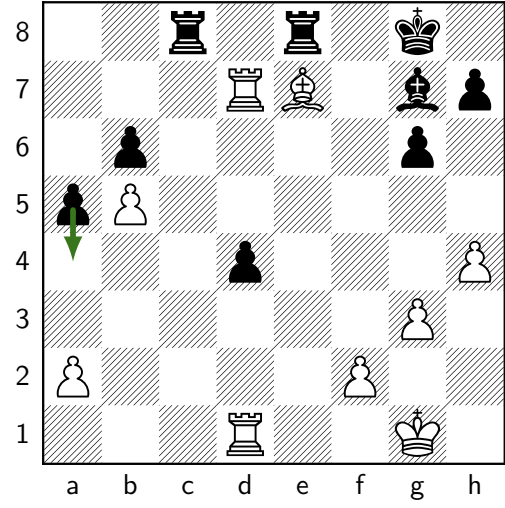


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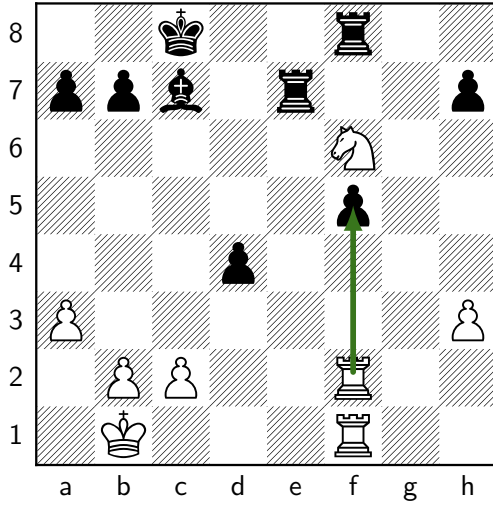
Figure 20: **Move 26, Black: Bb7 (c8b7)**. (a) Our position; Melanie plays Bb7 (c8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2271` (ply 49) [structural distance  $d=55$ , QAP obj=45]: the master played Rab8; under  $\mu$  our *Bc1* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *pg3* as their *Ba7*, our *rc7* as their *bb7*; transferred move c8b7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1607` (ply 55) [structural distance  $d=44$ , QAP obj=44]: the master played Rf5+; under  $\mu$  our *Bc1* plays the role of that game’s *Rd4* (a **cross-type** correspondence); cross-type roles: our *Pe5* as their *Bf8*, our *rc7* as their *nc7*; transferred move c8b7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2897` (ply 48) [structural distance  $d=60$ , QAP obj=38]: the master played Nd5; under  $\mu$  our *Bc1* plays the role of that game’s *Nc3* (a **cross-type** correspondence); cross-type roles: our *Pd7* as their *Rd1*, our *Rf8* as their *Pa2*; transferred move c8b7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bb7 (summed QAP objective 185, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



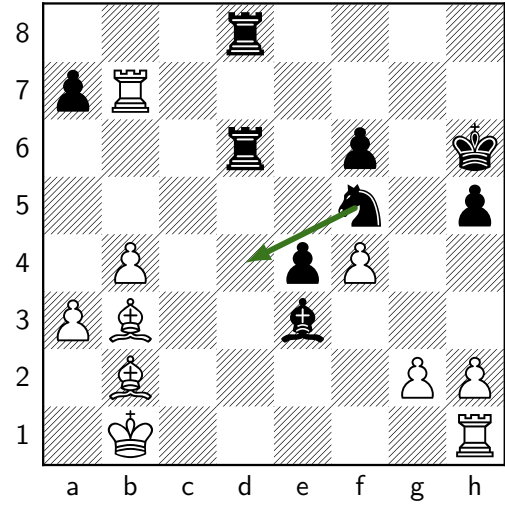
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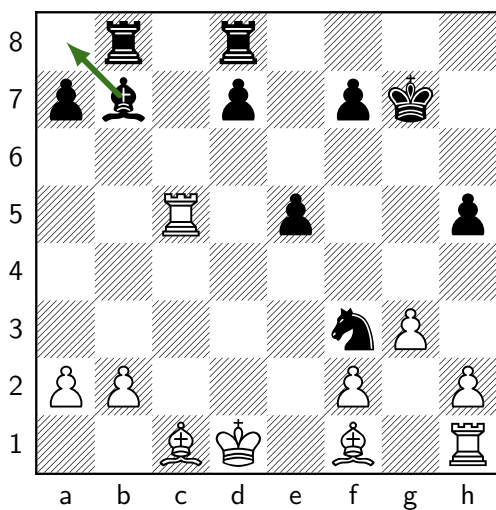


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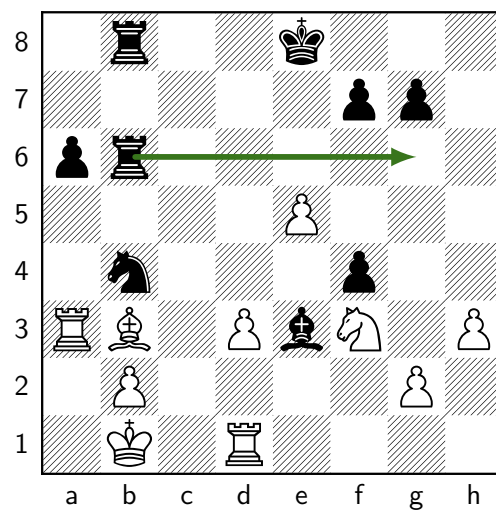


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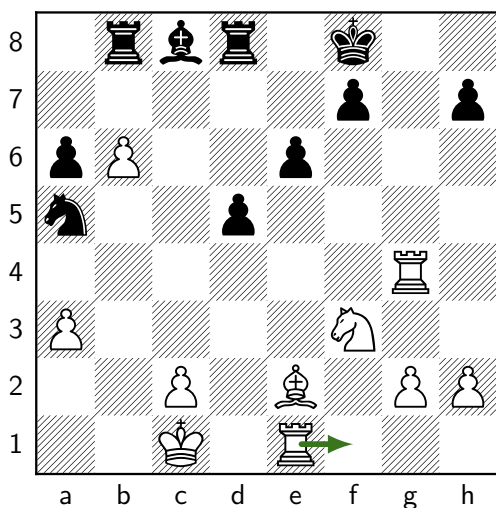
Figure 21: **Move 27, Black: Rfd8**. (a) Our position; Melanie plays Rfd8 (f8d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#841 (ply 53) [structural distance  $d=49$ , QAP obj=24]: the master played a4; under  $\mu$  our Rf1 plays the role of that game’s Pa4 (a **cross-type** correspondence); cross-type roles: our bf1 as their Rc8, our pf2 as their Bg7; transferred move f8d8 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0003#325 (ply 54) [structural distance  $d=54$ , QAP obj=14]: the master played Rxf5; under  $\mu$  our Rf1 plays the role of that game’s Rf2; cross-type roles: our Pd7 as their Rf1, our bc1 as their Pb2; transferred move f8d8 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#76 (ply 55) [structural distance  $d=45$ , QAP obj=10]: the master played Nd4; under  $\mu$  our Rf1 plays the role of that game’s Nf4 (a **cross-type** correspondence); cross-type roles: our pf2 as their Rd8, our Pe5 as their Rd6; transferred move f8d8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rfd8 (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



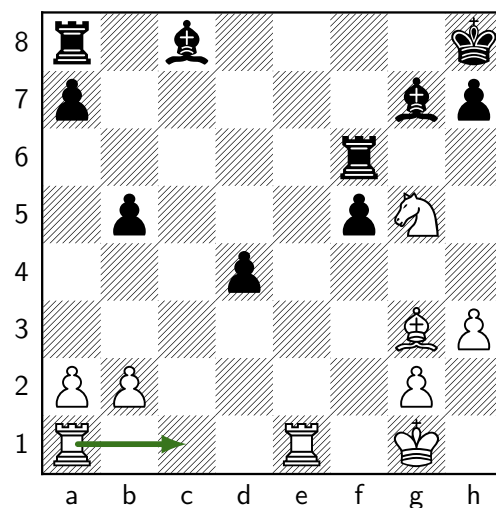
(a)



(b)



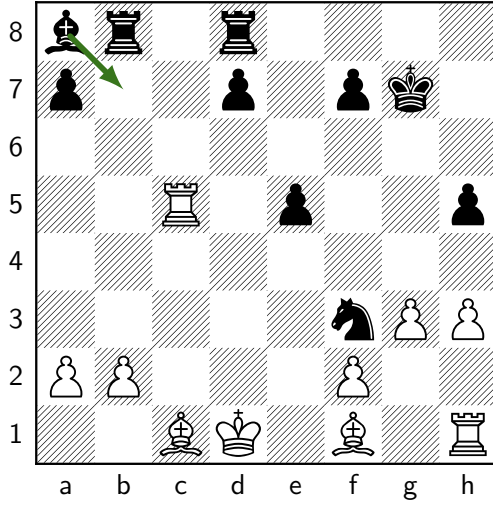
(c)



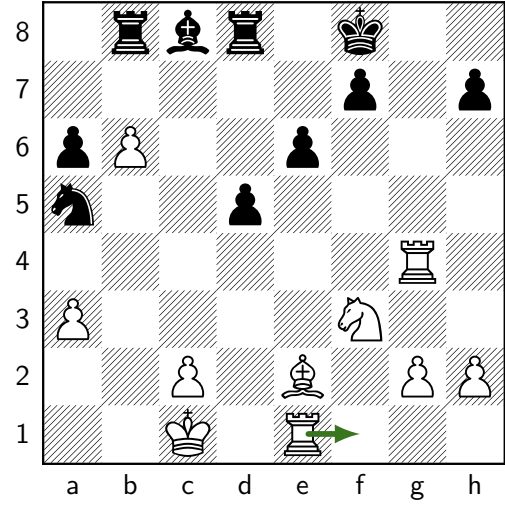
(d)

Figure 22: **Move 28, Black: Ba8**. (a) Our position; Melanie plays Ba8 (b7a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#695` (ply 65) [structural distance  $d=44$ , QAP obj=35]: the master played Rg6; under  $\mu$  our Bb2 plays the role of that game’s Rb3 (a **cross-type** correspondence); cross-type roles: our bf1 as their Rb8, our Rd8 as their Pf7; transferred move b7a8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1328` (ply 58) [structural distance  $d=52$ , QAP obj=29]: the master played Rf1; under  $\mu$  our Bb2 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Be2, our Rb8 as their Pg2; transferred move b7a8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1191` (ply 52) [structural distance  $d=43$ , QAP obj=24]: the master played Rac1; under  $\mu$  our Bb2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Ph5 as their Re1, our rc5 as their Pb2; transferred move b7a8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ba8 (summed QAP objective 88, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

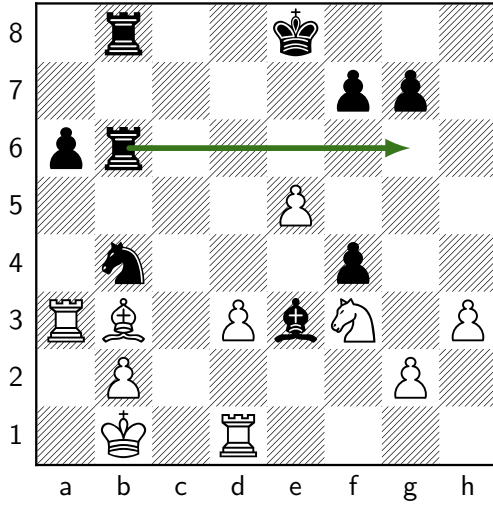




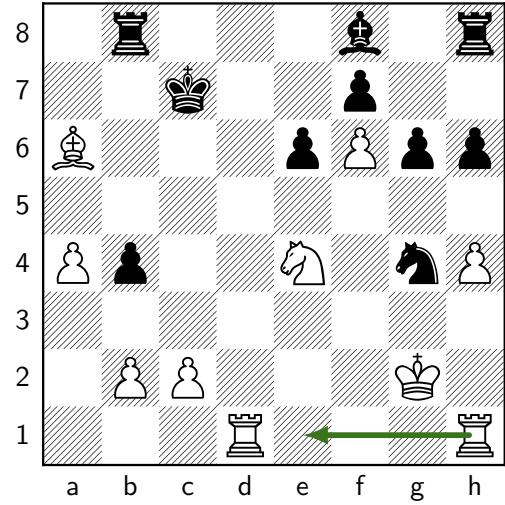
(a)



(b)

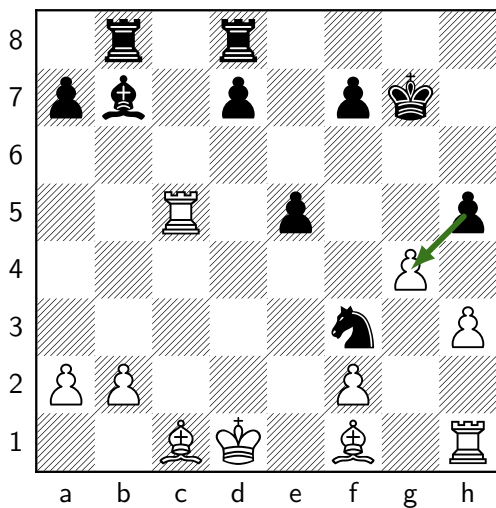


(c)

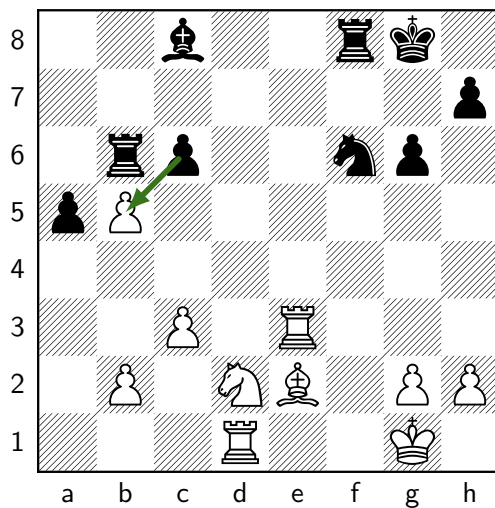


(d)

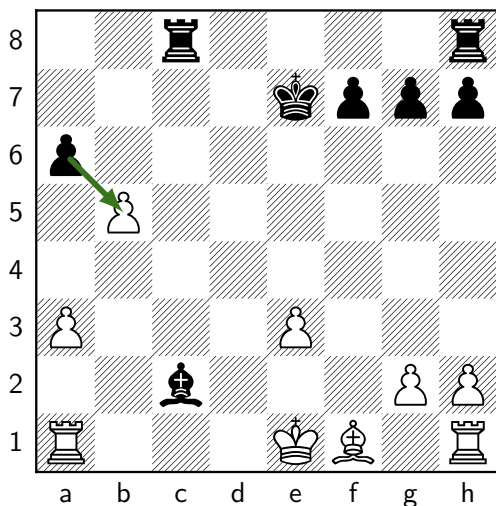
Figure 23: **Move 29, Black: Bb7.** (a) Our position; Melanie plays Bb7 (a8b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1328` (ply 58) [structural distance  $d=41$ , QAP obj=27]: the master played Rf1; under  $\mu$  our *Ba1* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Rb8* as their *Be2*, our *bf1* as their *Pg2*; transferred move a8b7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#695` (ply 65) [structural distance  $d=48$ , QAP obj=27]: the master played Rg6; under  $\mu$  our *Ba1* plays the role of that game’s *Rb3* (a **cross-type** correspondence); cross-type roles: our *bc1* as their *Pa6*, our *rc5* as their *Nb4*; transferred move a8b7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3914` (ply 58) [structural distance  $d=50$ , QAP obj=23]: the master played Rhe1; under  $\mu$  our *Ba1* plays the role of that game’s *Rh1* (a **cross-type** correspondence); cross-type roles: our *rc5* as their *ng4*, our *Rd8* as their *Ba6*; transferred move a8b7 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Bb7 (summed QAP objective 145, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



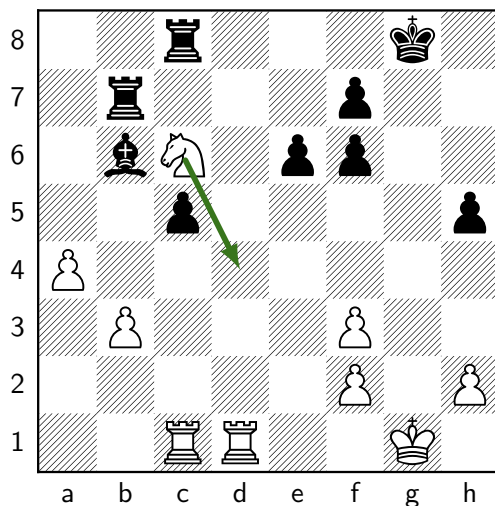
(a)



(b)

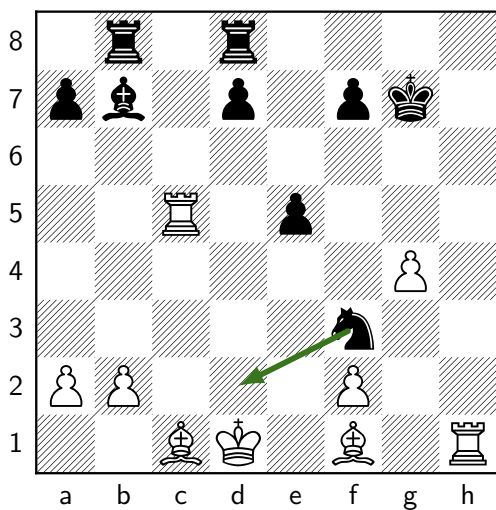


(c)

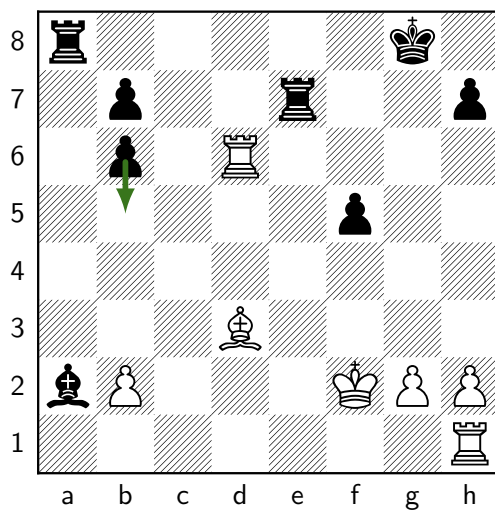


(d)

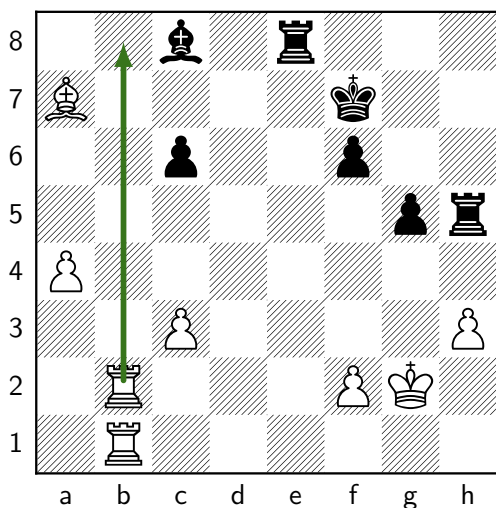
Figure 24: **Move 30, Black: hxg4.** (a) Our position; Melanie plays hxg4 (h5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#1538 (ply 45) [structural distance  $d=54$ , QAP obj=51]: the master played cxb5; under  $\mu$  our Ph4 plays the role of that game’s Pc3; cross-type roles: our Nf3 as their Bc8, our Ph5 as their Rb6; transferred move h5g4 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0001#1681 (ply 43) [structural distance  $d=41$ , QAP obj=45]: the master played axb5; under  $\mu$  our Ph4 plays the role of that game’s Pa3; cross-type roles: our Bb7 as their Rc8, our Pa7 as their Rh8; transferred move h5g4 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0002#785 (ply 54) [structural distance  $d=53$ , QAP obj=15]: the master played Nd4; under  $\mu$  our Ph4 plays the role of that game’s Nc6 (a **cross-type** correspondence); cross-type roles: our Nf3 as their Rd1, our bf1 as their Pf2; transferred move h5g4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to hxg4 (summed QAP objective 111, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



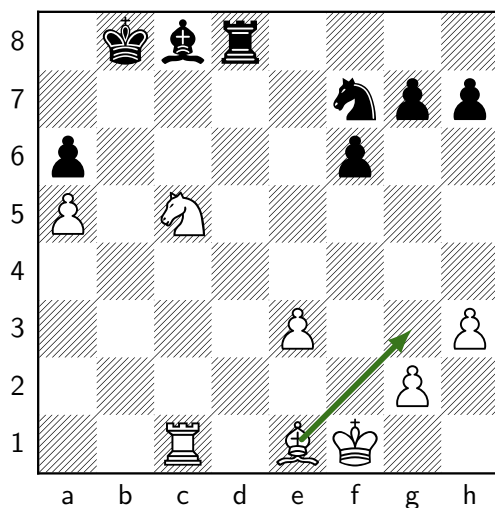
(a)



(b)



(c)



(d)

Figure 25: **Move 31, Black: Nd2.** (a) Our position; Melanie plays Nd2 (f3d2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2581` (ply 53) [structural distance  $d=60$ , QAP obj=27]: the master played **b5**; under  $\mu$  our *Pe4* plays the role of that game’s *Pb3*; cross-type roles: our *Nf3* as their *Ra8*, our *bc1* as their *Pb7*; transferred move **e5e4**. (c) Retrieved from `ccr1_top_p0000#2221` (ply 74) [structural distance  $d=58$ , QAP obj=25]: the master played **Rb8**; under  $\mu$  our *Rd1* plays the role of that game’s *Rb2*; cross-type roles: our *bc1* as their *pg5*, our *Nf3* as their *pf6*; transferred move **d8c8**. (d) Retrieved from `ccr1_top_p0000#278` (ply 68) [structural distance  $d=55$ , QAP obj=24]: the master played **Bg3+**; under  $\mu$  our *Pf2* plays the role of that game’s *Be1* (a **cross-type** correspondence); cross-type roles: our *Bb7* as their *Rc1*, our *Nf3* as their *Ph3*; transferred move **f7f5**. Synthesis: no retrieved analog transferred to Nd2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

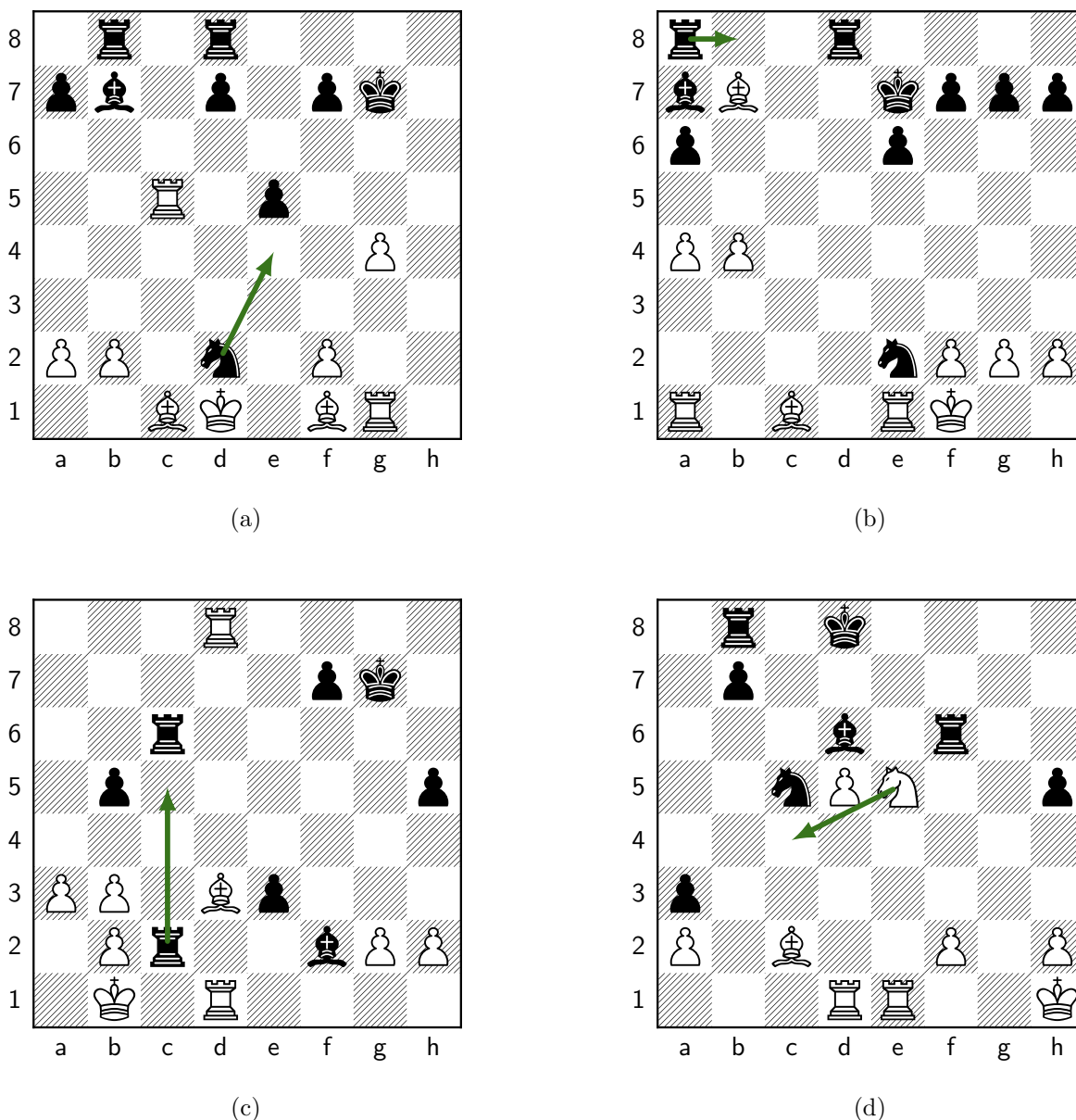
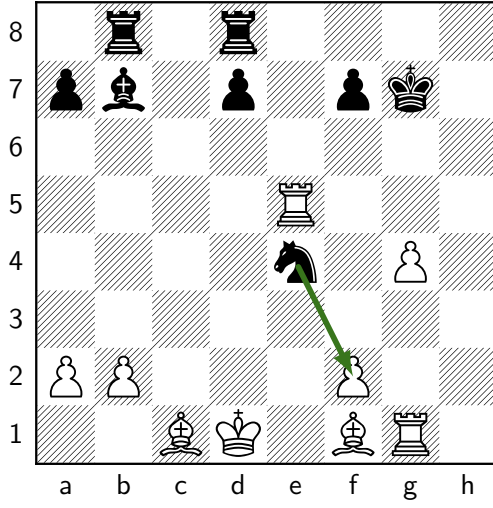
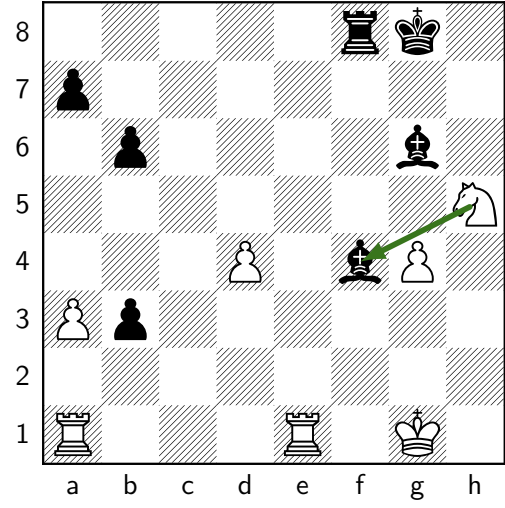


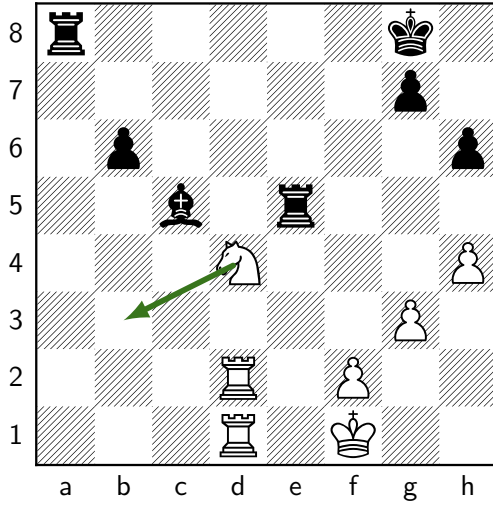
Figure 26: **Move 32, Black: Ne4**. (a) Our position; Melanie plays Ne4 (d2e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2271` (ply 49) [structural distance  $d=59$ , QAP obj=68]: the master played Rab8; under  $\mu$  our Nd7 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Pe5 as their Rd8, our Rd8 as their Ph7; transferred move d2e4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1703` (ply 69) [structural distance  $d=64$ , QAP obj=44]: the master played R2c5; under  $\mu$  our Nd7 plays the role of that game’s Rc7 (a **cross-type** correspondence); cross-type roles: our Bb7 as their Rc6, our Rd8 as their Ph5; transferred move d2e4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1021` (ply 66) [structural distance  $d=59$ , QAP obj=24]: the master played Nc4; under  $\mu$  our Nd7 plays the role of that game’s Ne5; cross-type roles: our Rb8 as their Pa2, our Pa7 as their Bc2; transferred move d2e4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ne4 (summed QAP objective 154, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



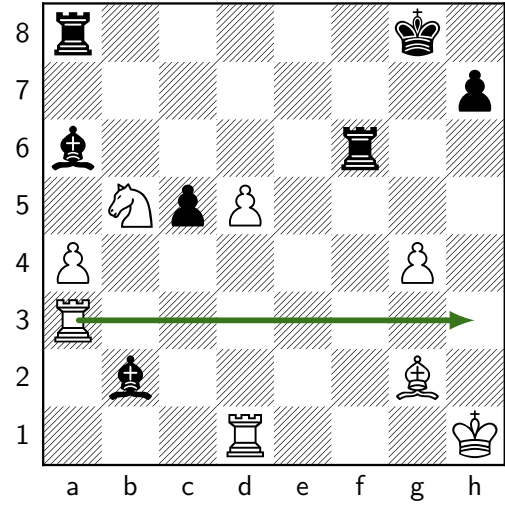
(a)



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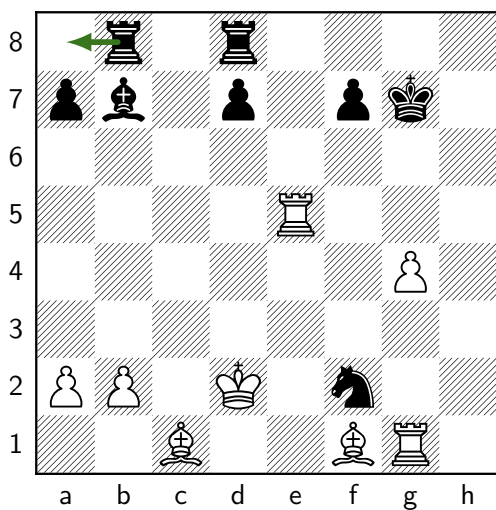


(c)

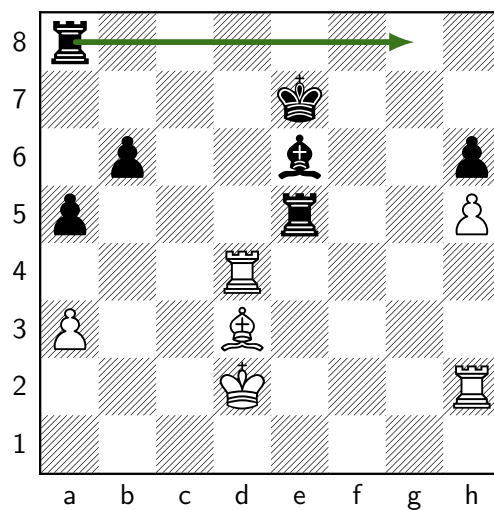


(d)

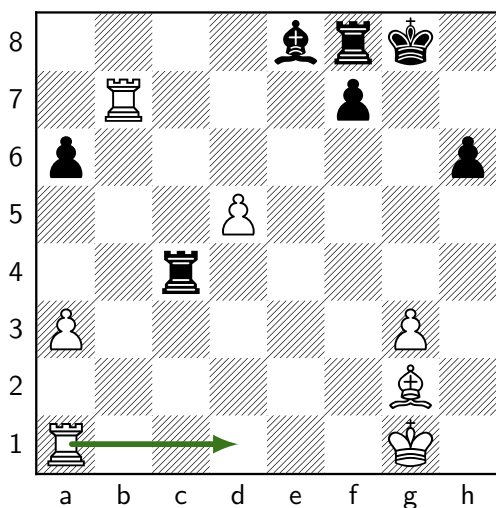
Figure 27: **Move 33, Black: Nxf2+.** (a) Our position; Melanie plays Nxf2+ (e4f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2790` (ply 78) [structural distance  $d=49$ , QAP obj=25]: the master played Nxf4; under  $\mu$  our Ne5 plays the role of that game’s Nh5; cross-type roles: our Pf7 as their Re1, our Bb7 as their pb3; transferred move e4f2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#80` (ply 68) [structural distance  $d=50$ , QAP obj=20]: the master played Nb3; under  $\mu$  our Ne5 plays the role of that game’s Nd4; cross-type roles: our pa2 as their Rd1, our Bb7 as their Rd2; transferred move e4f2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#2828` (ply 72) [structural distance  $d=52$ , QAP obj=14]: the master played Rh3; under  $\mu$  our Ne5 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our Pd7 as their Rd1, our re5 as their bb2; transferred move e4f2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Nxf2+ (summed QAP objective 74, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



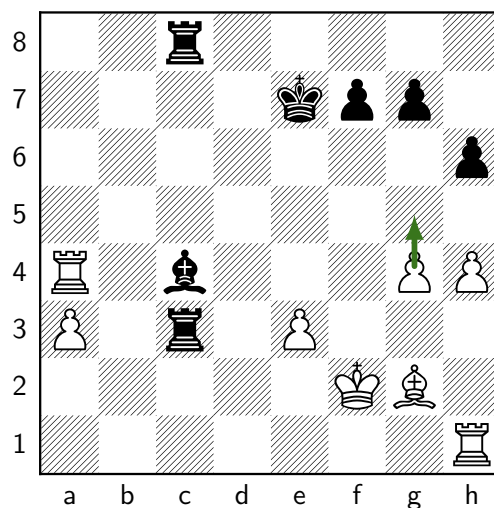
(a)



(b)

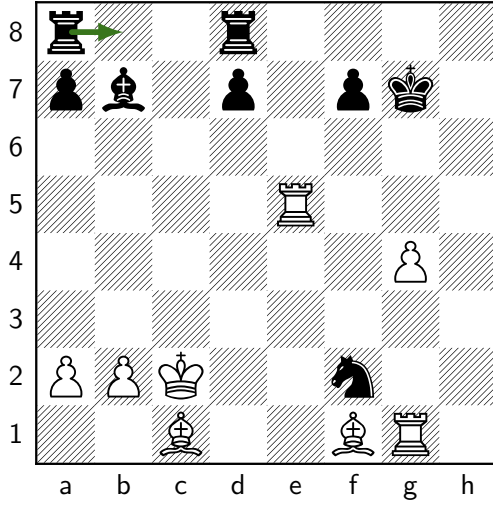


(c)

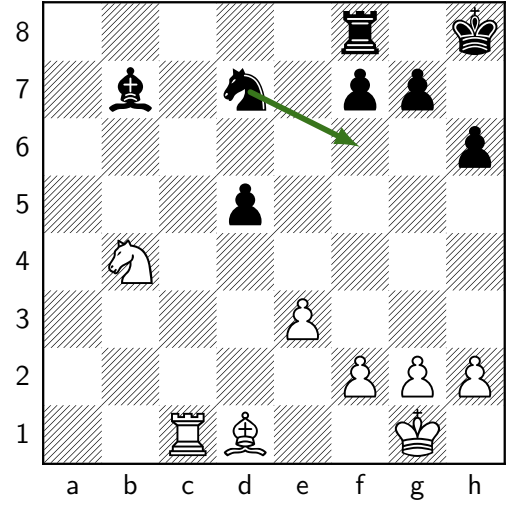


(d)

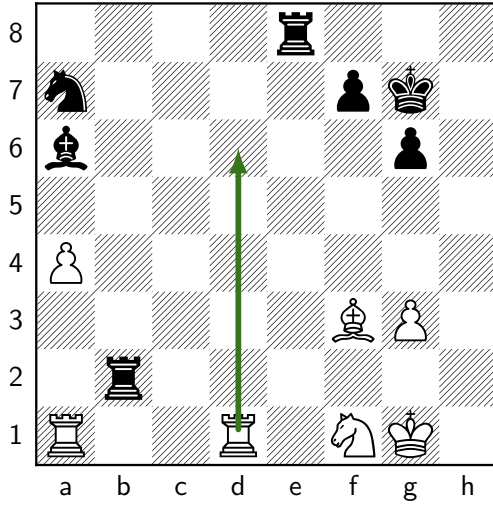
Figure 28: **Move 34, Black: Ra8 (b8a8)**. (a) Our position; Melanie plays Ra8 (b8a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2247` (ply 69) [structural distance  $d=52$ , QAP obj=16]: the master played Rg8; under  $\mu$  our Rb1 plays the role of that game’s Ra1; cross-type roles: our pb2 as their Be6, our Bb7 as their Ph6; transferred move b8a8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#261` (ply 64) [structural distance  $d=51$ , QAP obj=13]: the master played Rd1; under  $\mu$  our Rb1 plays the role of that game’s Ra1; cross-type roles: our Rd8 as their Bg2, our Bb7 as their Pd5; transferred move b8a8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3139` (ply 64) [structural distance  $d=50$ , QAP obj=10]: the master played g5; under  $\mu$  our Rb1 plays the role of that game’s Pg4 (a **cross-type** correspondence); cross-type roles: our Pf7 as their Rh1, our Nf2 as their rc3; transferred move b8a8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ra8 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



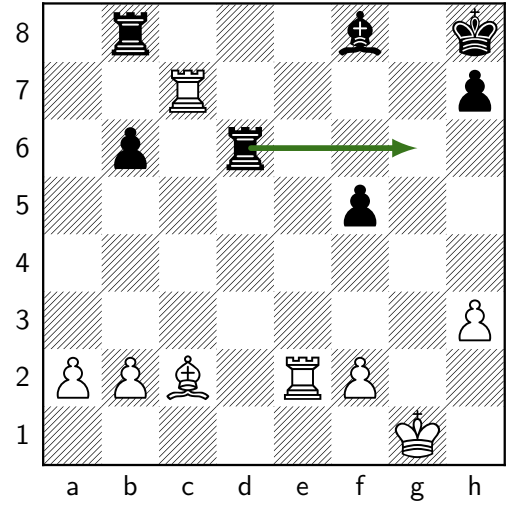
(a)



(b)



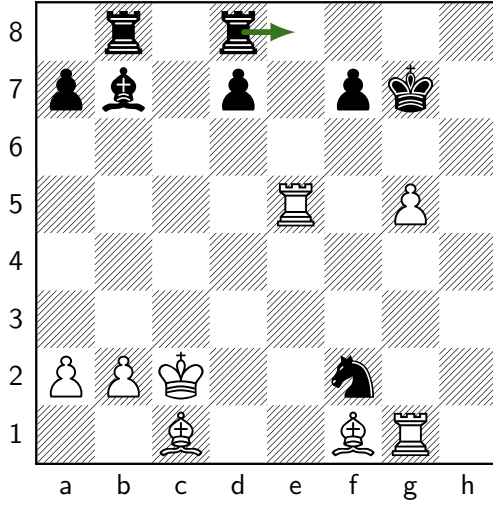
(c)



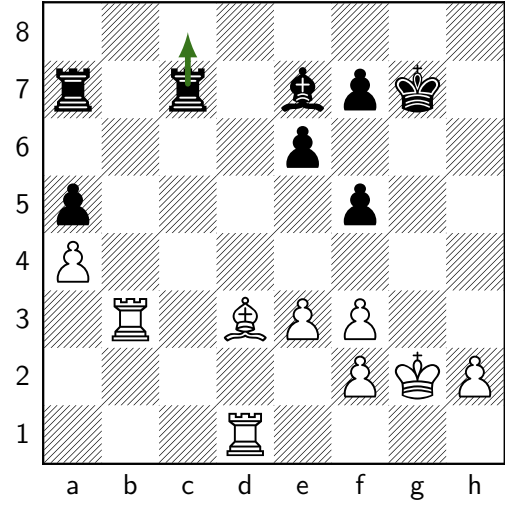
(d)

Figure 29: **Move 35, Black: Rab8.** (a) Our position; Melanie plays Rab8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3098` (ply 59) [structural distance  $d=51$ , QAP obj=20]: the master played Nf6; under  $\mu$  our *Ra1* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *rg1* as their *Bb7*, our *bf1* as their *Pf7*; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1598` (ply 78) [structural distance  $d=47$ , QAP obj=19]: the master played Rd6; under  $\mu$  our *Ra1* plays the role of that game’s *Rd1*; cross-type roles: our *pg4* as their *Nf1*, our *bf1* as their *Pa4*; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1906` (ply 71) [structural distance  $d=42$ , QAP obj=16]: the master played Rg6+; under  $\mu$  our *Ra1* plays the role of that game’s *Rd3*; cross-type roles: our *Rd8* as their *ph3*, our *Nf2* as their *bc2*; transferred move a8b8 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Rab8 (summed QAP objective 111, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

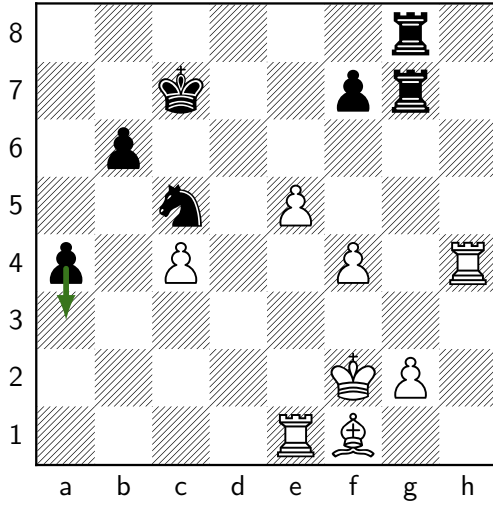




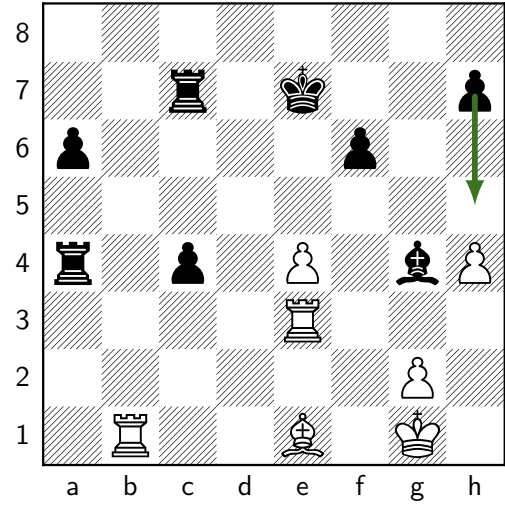
(a)



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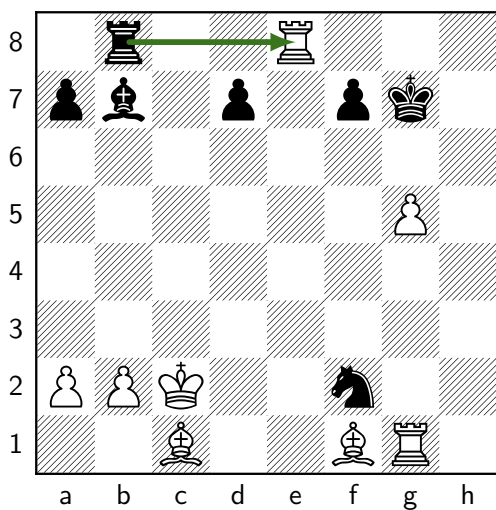


(c)

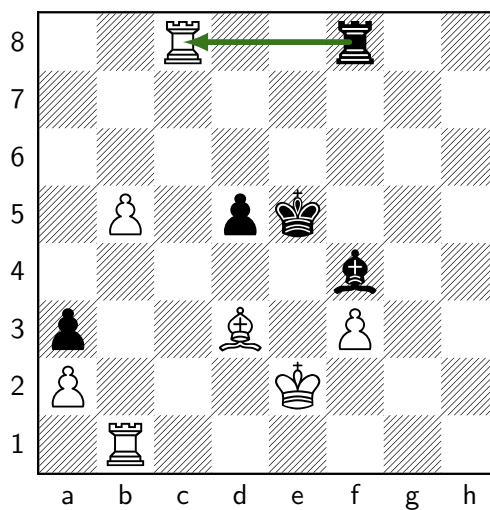


(d)

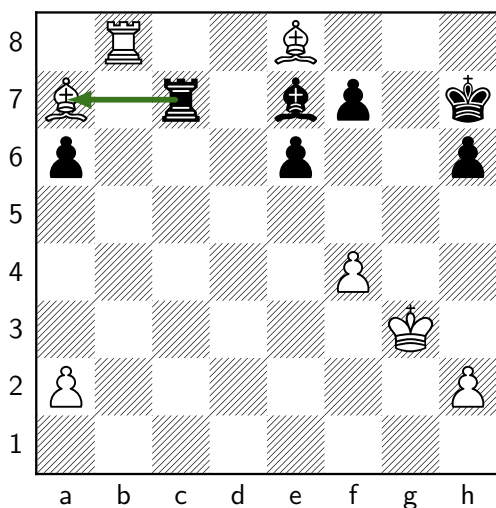
Figure 30: **Move 36, Black: Re8**. (a) Our position; Melanie plays Re8 (d8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1983` (ply 57) [structural distance  $d=56$ , QAP obj=6]: the master played Rc8; under  $\mu$  our *Rd1* plays the role of that game’s *Rc2*; cross-type roles: our *Pd7* as their *Be7*, our *Nf2* as their *Pe6*; transferred move d8e8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2853` (ply 71) [structural distance  $d=34$ , QAP obj=5]: the master played a3; under  $\mu$  our *Rd1* plays the role of that game’s *Pa5* (a **cross-type** correspondence); cross-type roles: our *bf1* as their *Rg8*, our *Rb8* as their *Nc5*; transferred move d8e8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#367` (ply 77) [structural distance  $d=48$ , QAP obj=3]: the master played h5; under  $\mu$  our *Rd1* plays the role of that game’s *Ph2* (a **cross-type** correspondence); cross-type roles: our *Bb7* as their *Ra4*, our *Nf2* as their *Pc4*; transferred move d8e8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Re8 (summed QAP objective 17, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



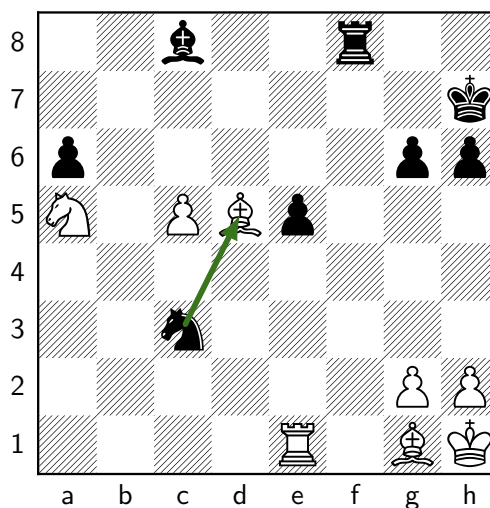
(a)



(b)

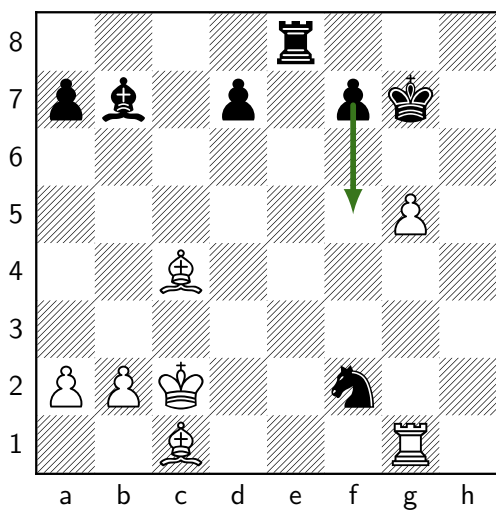


(c)

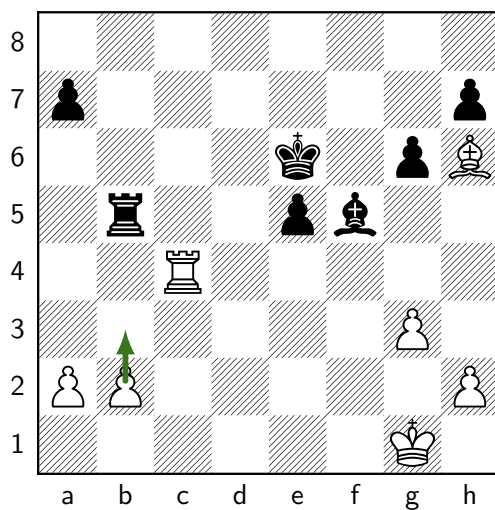


(d)

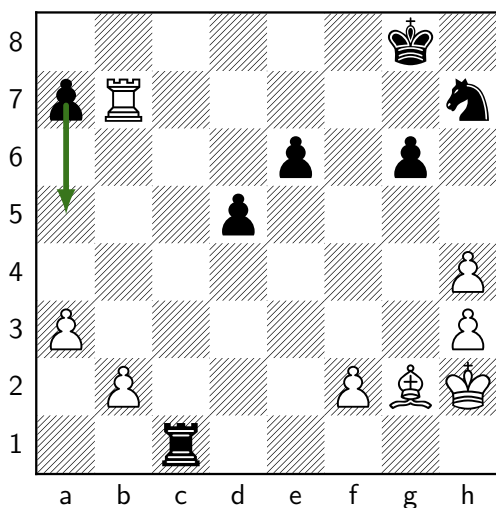
Figure 31: **Move 37, Black: Rxe8**. (a) Our position; Melanie plays Rxe8 (b8e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#2144 (ply 87) [structural distance  $d=38$ , QAP obj=43]: the master played Rxc8; under  $\mu$  our Rb1 plays the role of that game’s Rf1; cross-type roles: our Nf2 as their Pa3, our Pa7 as their bd3; transferred move b8e8 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#3741 (ply 69) [structural distance  $d=40$ , QAP obj=20]: the master played Rxa7; under  $\mu$  our Rb1 plays the role of that game’s Rc2; cross-type roles: our re8 as their be8, our re8 as their ba7; transferred move b8e8 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#398 (ply 75) [structural distance  $d=39$ , QAP obj=20]: the master played Nxd5; under  $\mu$  our Rb1 plays the role of that game’s Nc6 (a **cross-type** correspondence); cross-type roles: our Pd7 as their Rf8, our pb2 as their na5; transferred move b8e8 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Rxe8 (summed QAP objective 185, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



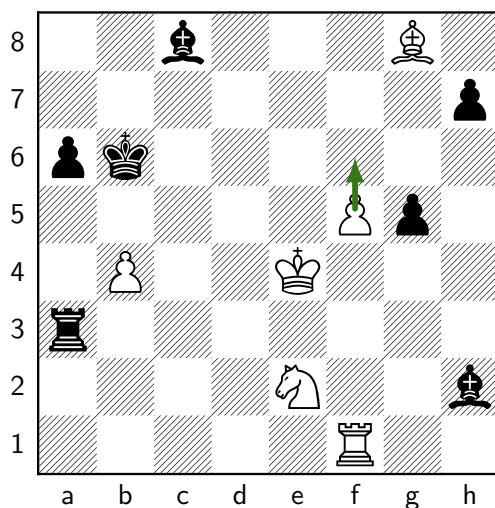
(a)



(b)



(c)



(d)

Figure 32: **Move 38, Black: f5 (f7f5)**. (a) Our position; Melanie plays f5 (f7f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1461` (ply 64) [structural distance  $d=39$ , QAP obj=20]: the master played b3; under  $\mu$  our  $Pf2$  plays the role of that game’s  $Pb2$ ; cross-type roles: our  $bc1$  as their  $Pg3$ , our  $bc4$  as their  $rb5$ ; transferred move f7f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1039` (ply 73) [structural distance  $d=43$ , QAP obj=15]: the master played a5; under  $\mu$  our  $Pf2$  plays the role of that game’s  $Pa2$ ; cross-type roles: our  $bc4$  as their  $rb7$ , our  $Pf7$  as their  $Nh7$ ; transferred move f7f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#163` (ply 82) [structural distance  $d=41$ , QAP obj=10]: the master played f6; under  $\mu$  our  $Pf2$  plays the role of that game’s  $Pf5$ ; cross-type roles: our  $bc1$  as their  $Rf1$ , our  $Pa7$  as their  $Ne2$ ; transferred move f7f5 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to f5 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

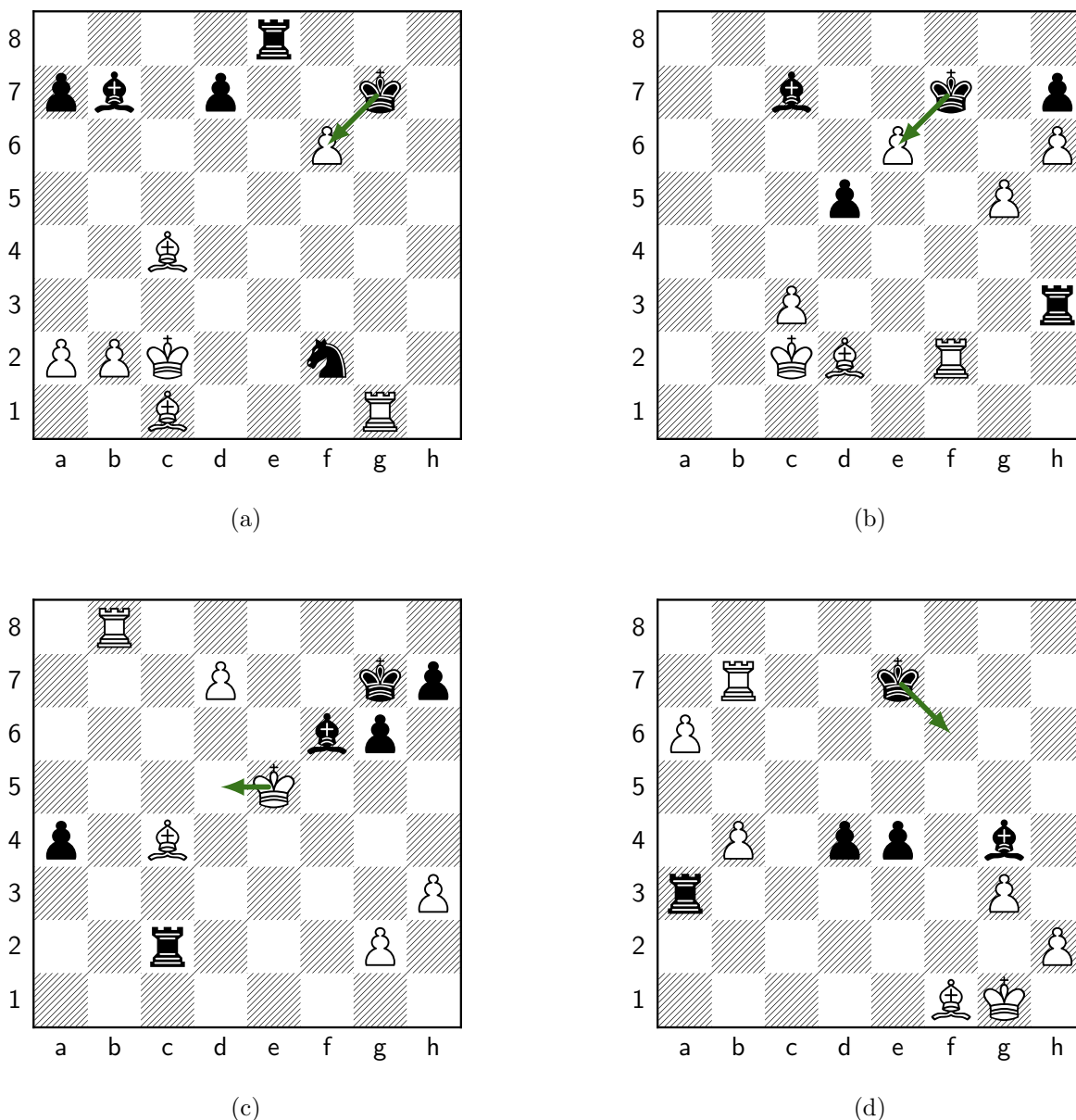
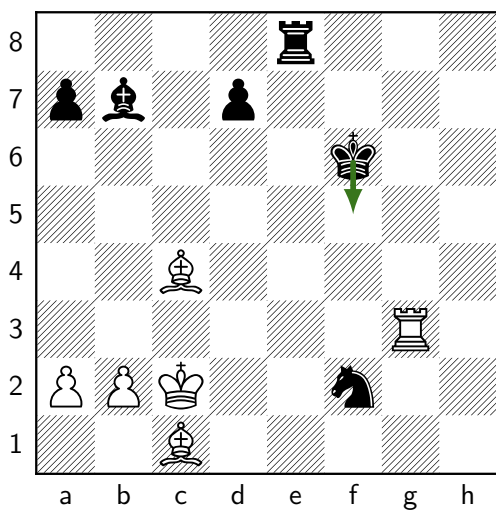
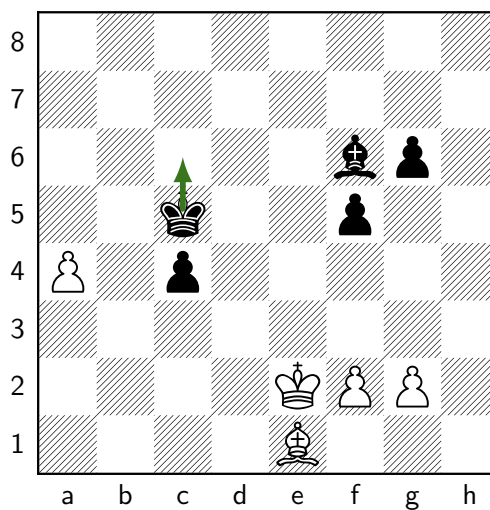


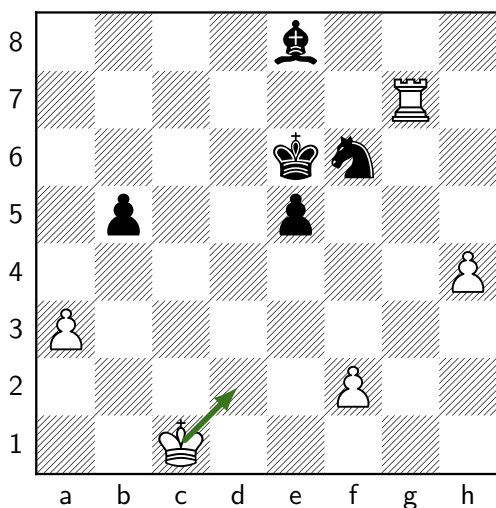
Figure 33: **Move 39, Black: Kxf6.** (a) Our position; Melanie plays Kxf6 (g7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0001#1753 (ply 93) [structural distance  $d=39$ , QAP obj=84]: the master played Kxe6; under  $\mu$  our Kg2 plays the role of that game's Kf2; cross-type roles: our bc4 as their ph6, our Re8 as their Pd5; transferred move g7f6 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0002#2558 (ply 90) [structural distance  $d=35$ , QAP obj=67]: the master played Kd5; under  $\mu$  our Kg2 plays the role of that game's Ke5; cross-type roles: our pf6 as their bf6, our rg1 as their pg6; transferred move g7f6 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0000#1937 (ply 87) [structural distance  $d=30$ , QAP obj=25]: the master played Kf6; under  $\mu$  our Kg2 plays the role of that game's Ke2; cross-type roles: our pf6 as their rb7, our Bb7 as their pb4; transferred move g7f6 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kxf6 (summed QAP objective 247, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



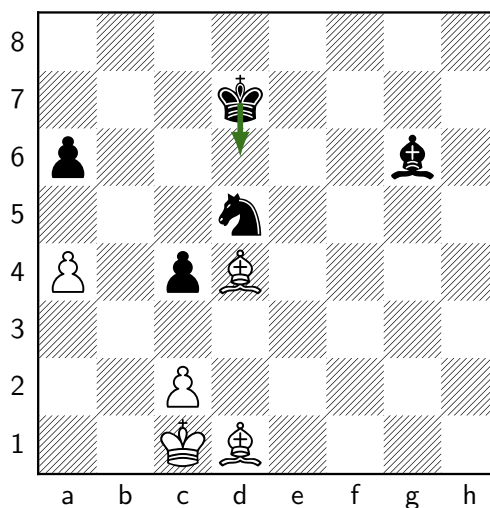
(a)



(b)

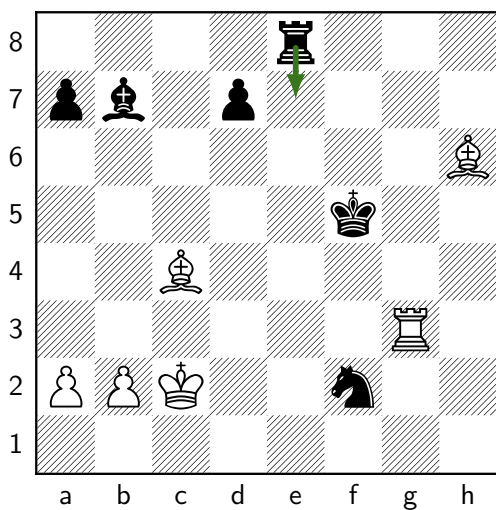


(c)

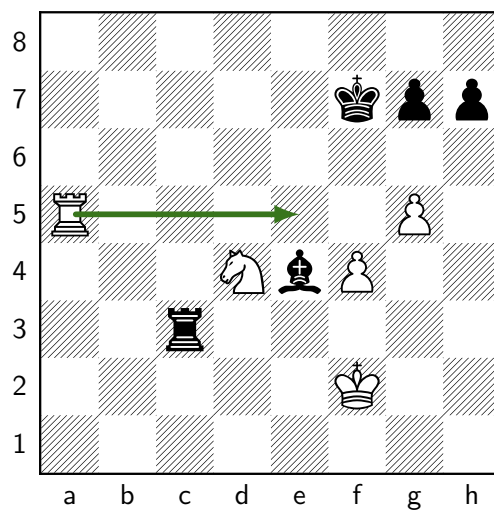


(d)

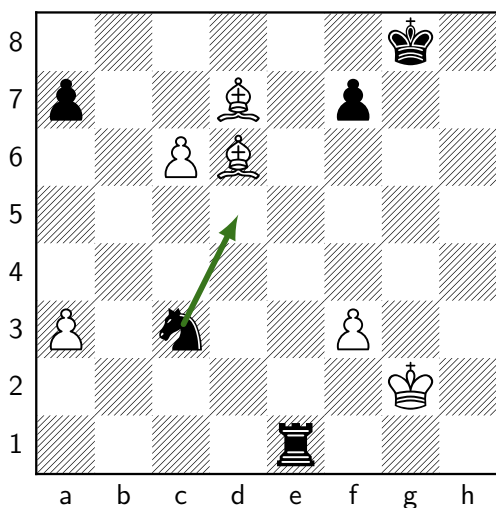
Figure 34: **Move 40, Black: Kf5.** (a) Our position; Melanie plays Kf5 (f6f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#972` (ply 95) [structural distance  $d=29$ , QAP obj=5]: the master played Kc6; under  $\mu$  our Kf3 plays the role of that game’s Kc4; cross-type roles: our Re8 as their Bf6, our bc4 as their Pf5; transferred move f6f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3973` (ply 80) [structural distance  $d=27$ , QAP obj=4]: the master played Kd2; under  $\mu$  our Kf3 plays the role of that game’s Kc1; cross-type roles: our Re8 as their Pf2, our rg3 as their Pa3; transferred move f6f5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1766` (ply 91) [structural distance  $d=29$ , QAP obj=4]: the master played Kd6; under  $\mu$  our Kf3 plays the role of that game’s Kd2; cross-type roles: our rg3 as their Pa6, our Nf2 as their Bg6; transferred move f6f5 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Kf5 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



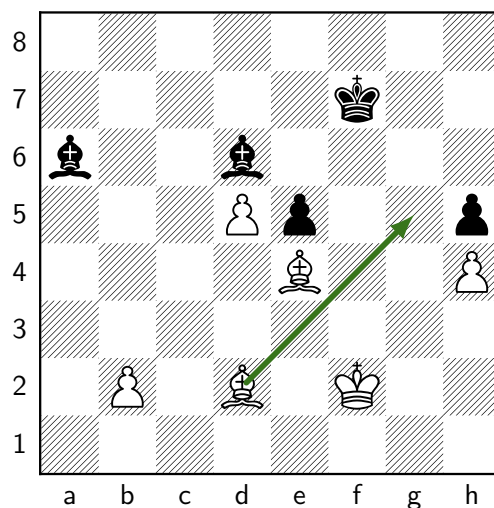
(a)



(b)

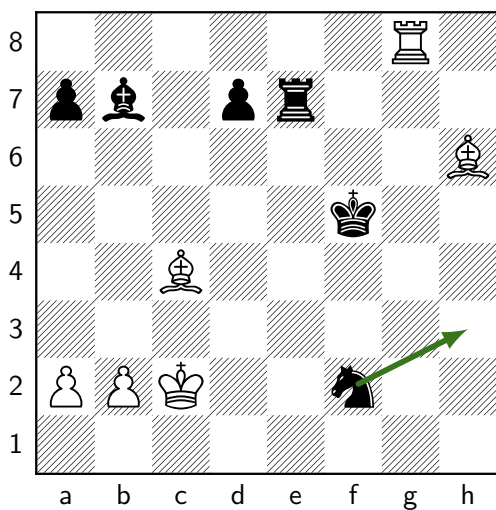


(c)

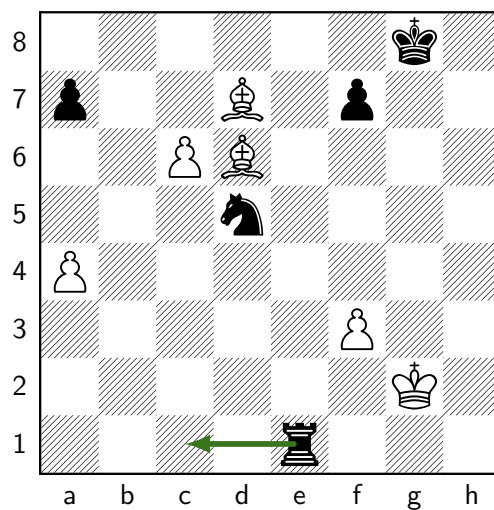


(d)

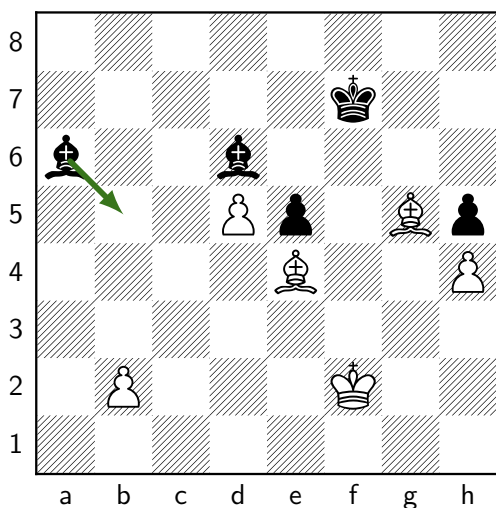
Figure 35: **Move 41, Black: Re7.** (a) Our position; Melanie plays **Re7** (**e8e7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#760` (ply 78) [structural distance  $d=24$ , QAP obj=5]: the master played **Re5**; under  $\mu$  our *Re1* plays the role of that game’s *Ra5*; cross-type roles: our *Bb7* as their *Nd4*, our *Nf2* as their *Pg5*; transferred move **e8e7** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3028` (ply 75) [structural distance  $d=21$ , QAP obj=4]: the master played **Nd5**; under  $\mu$  our *Re1* plays the role of that game’s *Nc6* (a **cross-type** correspondence); cross-type roles: our *Bb7* as their *Pa7*, our *rg3* as their *bd6*; transferred move **e8e7** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2815` (ply 92) [structural distance  $d=22$ , QAP obj=3]: the master played **Bg5**; under  $\mu$  our *Re1* plays the role of that game’s *Bd2* (a **cross-type** correspondence); cross-type roles: our *pb2* as their *Be4*, our *rg3* as their *Ph4*; transferred move **e8e7** (**endorses this move**). Synthesis: 9 retrieved analogs transferred to **Re7** (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



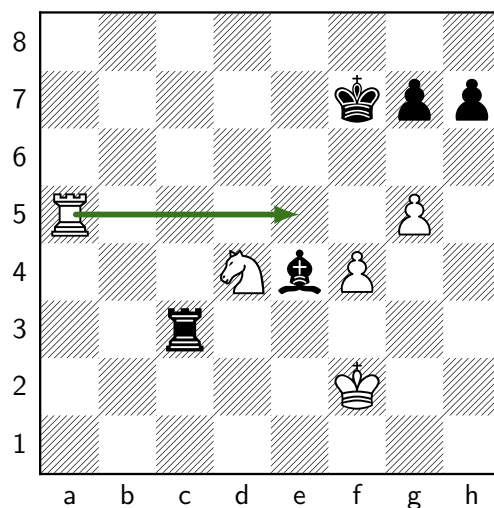
(a)



(b)



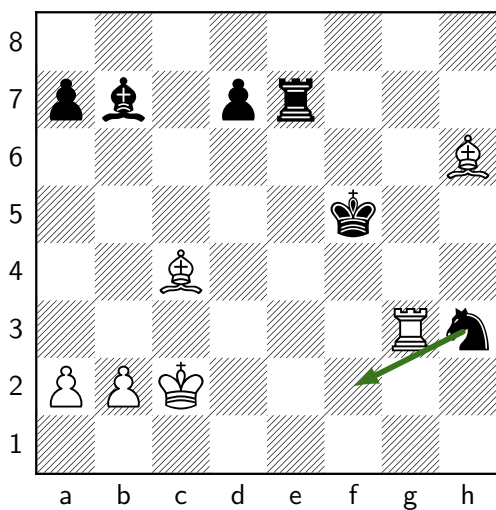
(c)



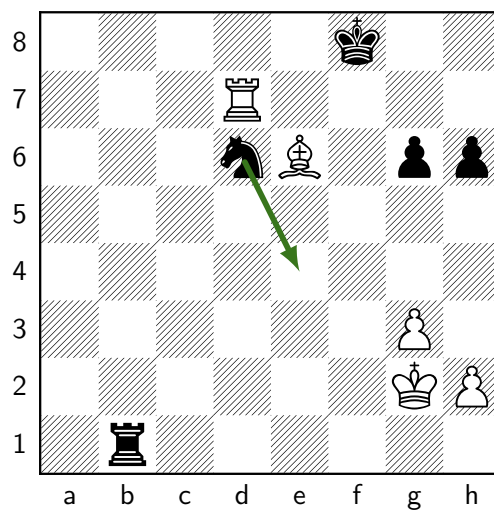
(d)

Figure 36: **Move 42, Black: Nh3**. (a) Our position; Melanie plays Nh3 (f2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3028` (ply 77) [structural distance  $d=19$ , QAP obj=5]: the master played Rc1; under  $\mu$  our Nf7 plays the role of that game’s Re8 (a **cross-type** correspondence); cross-type roles: our rg8 as their Pa7, our Re7 as their bd7; transferred move f2h3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2815` (ply 93) [structural distance  $d=26$ , QAP obj=5]: the master played Bb5; under  $\mu$  our Nf7 plays the role of that game’s Ba3 (a **cross-type** correspondence); cross-type roles: our Re7 as their Bd6, our bh6 as their pd5; transferred move f2h3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#760` (ply 78) [structural distance  $d=22$ , QAP obj=4]: the master played Re5; under  $\mu$  our Nf7 plays the role of that game’s Ra5 (a **cross-type** correspondence); cross-type roles: our Bb7 as their Nd4, our Re7 as their Pf4; transferred move f2h3 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Nh3 (summed QAP objective 33, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

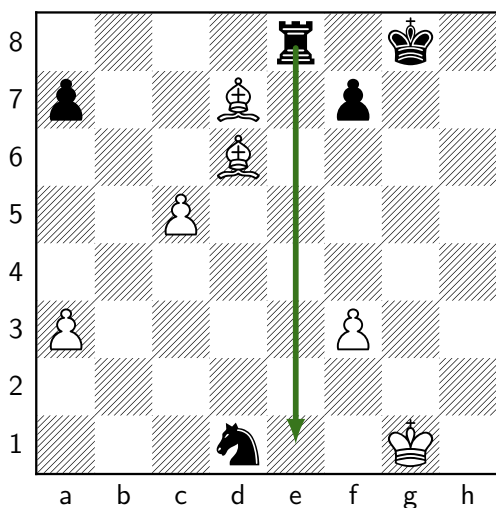




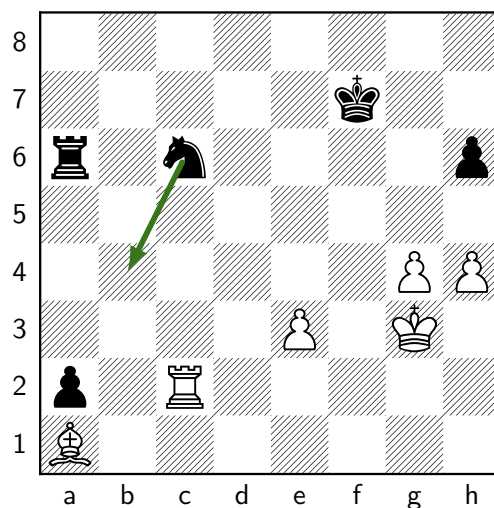
(a)



(b)

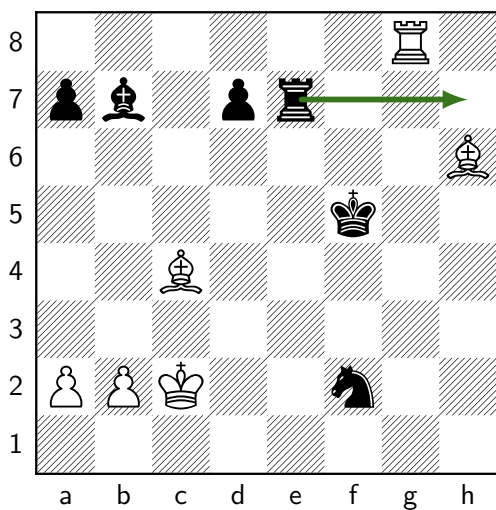


(c)

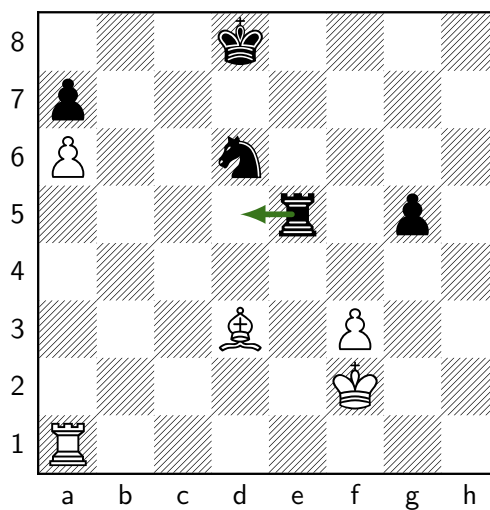


(d)

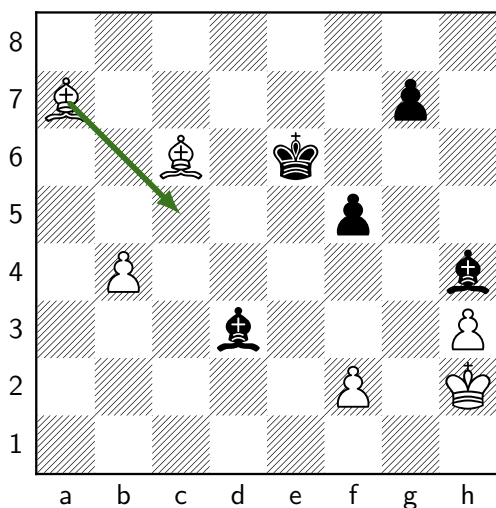
Figure 37: **Move 43, Black: Nf2**. (a) Our position; Melanie plays Nf2 (h3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3277` (ply 87) [structural distance  $d=20$ , QAP obj=21]: the master played Ne4; under  $\mu$  our Nh6 plays the role of that game’s Nd3; cross-type roles: our Re7 as their Pg6, our Bb7 as their Ph6; transferred move h3f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3028` (ply 71) [structural distance  $d=24$ , QAP obj=13]: the master played Re1+; under  $\mu$  our Nh6 plays the role of that game’s Re1 (a **cross-type** correspondence); cross-type roles: our bc4 as their Pa7, our rg3 as their bd7; transferred move h3f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#818` (ply 99) [structural distance  $d=24$ , QAP obj=8]: the master played Nb4; under  $\mu$  our Nh6 plays the role of that game’s Nc3; cross-type roles: our bc4 as their ph4, our bh6 as their pe3; transferred move h3f2 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Nf2 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



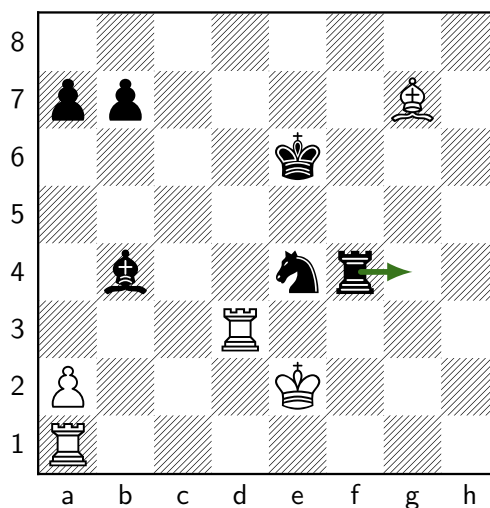
(a)



(b)

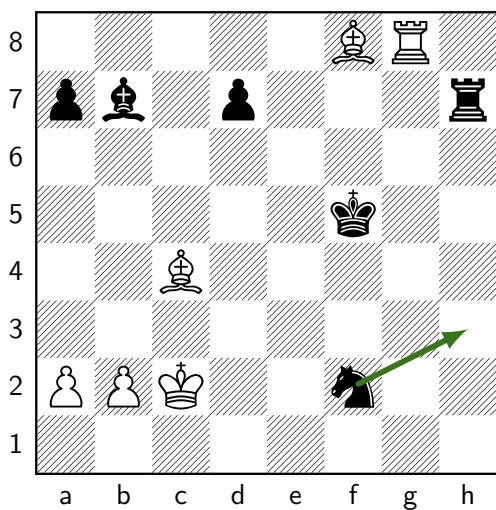


(c)

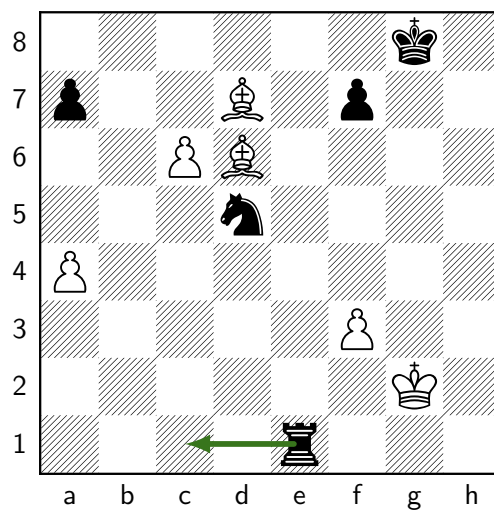


(d)

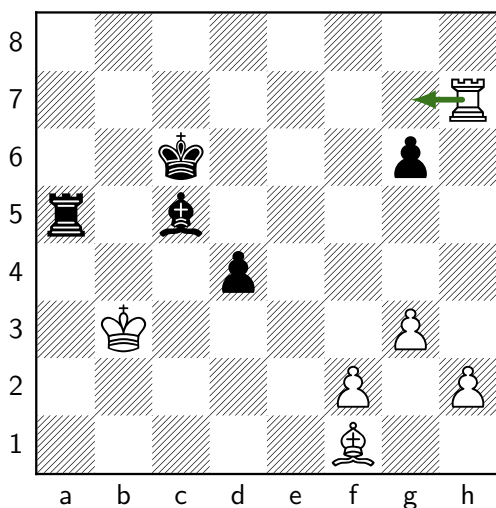
Figure 38: **Move 44, Black: Rh7.** (a) Our position; Melanie plays Rh7 (e7h7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0001#356` (ply 99) [structural distance  $d=25$ , QAP obj=5]: the master played Rd5; under  $\mu$  our Re2 plays the role of that game’s Re4; cross-type roles: our Bb7 as their Pa7, our rg8 as their pa6; transferred move e7h7 (**endorses this move**). (c) Retrieved from `cctl_top_p0001#225` (ply 92) [structural distance  $d=27$ , QAP obj=4]: the master played Bc5; under  $\mu$  our Re2 plays the role of that game’s Ba7 (a **cross-type** correspondence); cross-type roles: our Bb7 as their Pf2, our bh6 as their Ph3; transferred move e7h7 (**endorses this move**). (d) Retrieved from `cctl_top_p0001#2366` (ply 83) [structural distance  $d=22$ , QAP obj=4]: the master played Rg4; under  $\mu$  our Re2 plays the role of that game’s Rf5; cross-type roles: our rg8 as their Pa7, our bc4 as their rd3; transferred move e7h7 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Rh7 (summed QAP objective 40, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



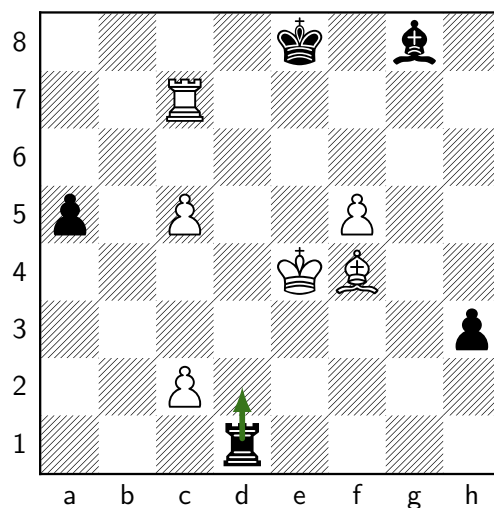
(a)



(b)

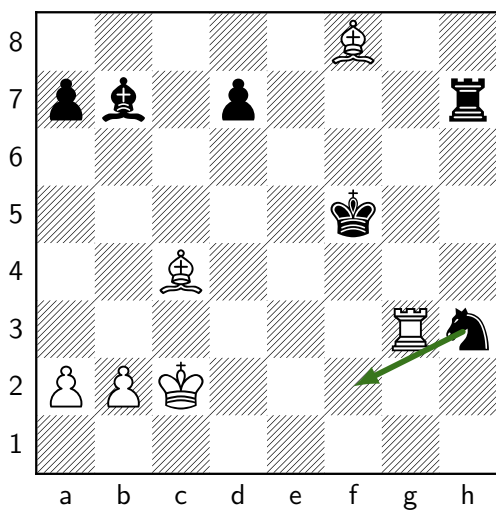


(c)

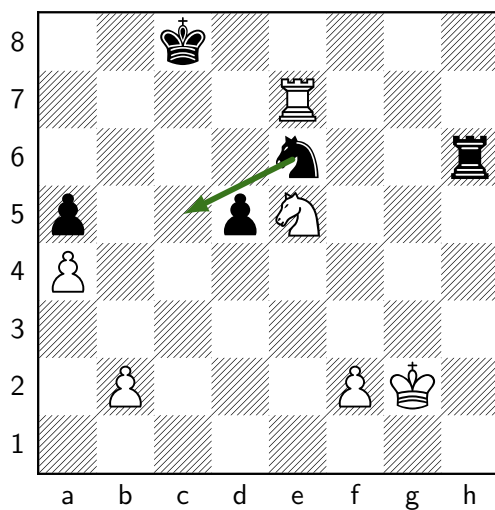


(d)

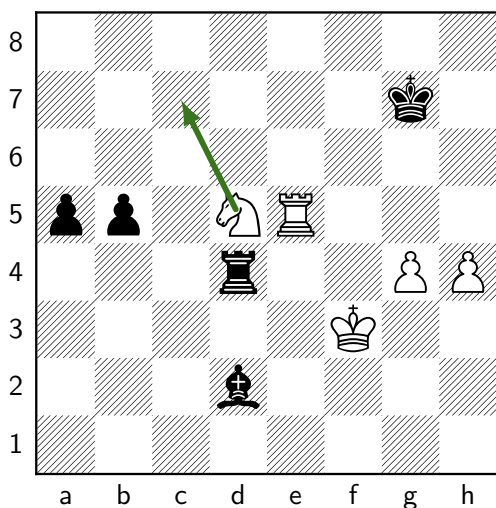
Figure 39: **Move 45, Black: Nh3**. (a) Our position; Melanie plays Nh3 (f2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#3028 (ply 77) [structural distance  $d=21$ , QAP obj=4]: the master played Rc1; under  $\mu$  our Nf7 plays the role of that game’s Re8 (a **cross-type** correspondence); cross-type roles: our bf8 as their Pa7, our Rh7 as their bd7; transferred move f2h3 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#1899 (ply 82) [structural distance  $d=27$ , QAP obj=3]: the master played Rg7; under  $\mu$  our Nf7 plays the role of that game’s Rh7 (a **cross-type** correspondence); cross-type roles: our Bb7 as their Pf2, our bc4 as their Pg3; transferred move f2h3 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#977 (ply 105) [structural distance  $d=18$ , QAP obj=3]: the master played Rd2; under  $\mu$  our Nf7 plays the role of that game’s Rd8 (a **cross-type** correspondence); cross-type roles: our Rh7 as their Bg8, our Bb7 as their Pa5; transferred move f2h3 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Nh3 (summed QAP objective 35, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



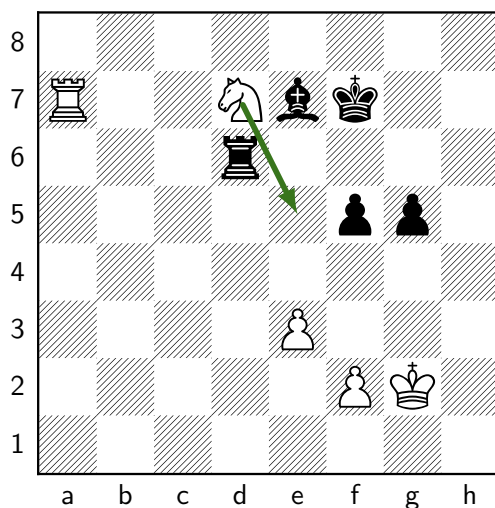
(a)



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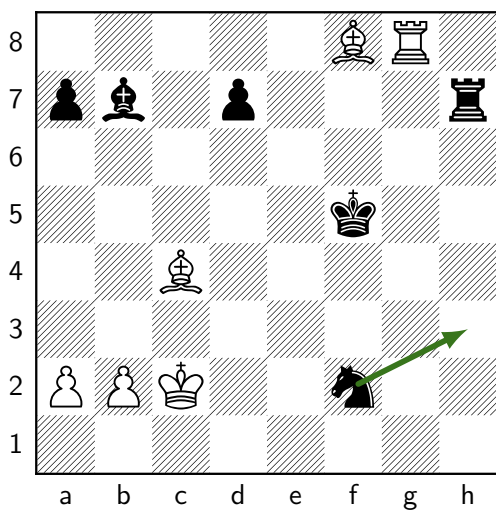


(c)

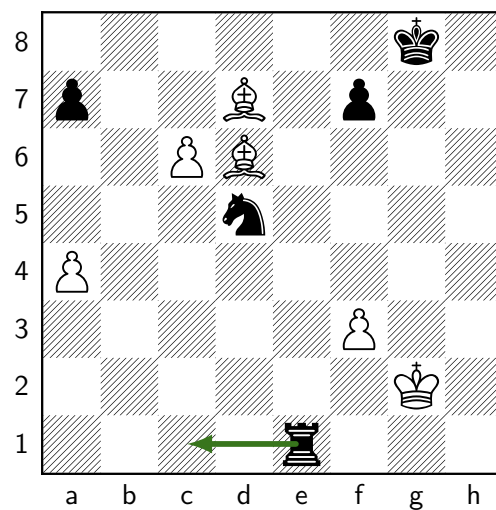


(d)

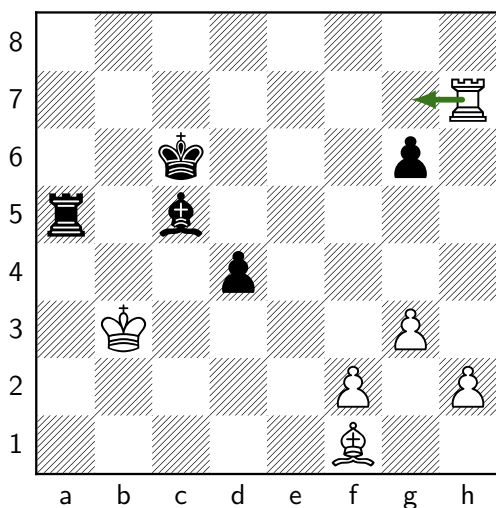
Figure 40: **Move 46, Black: Nf2**. (a) Our position; Melanie plays Nf2 (h3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1439` (ply 77) [structural distance  $d=22$ , QAP obj=25]: the master played Nc5; under  $\mu$  our Nh6 plays the role of that game’s Ne3; cross-type roles: our bf8 as their ne5, our bc4 as their pa4; transferred move h3f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2459` (ply 90) [structural distance  $d=25$ , QAP obj=23]: the master played Nc7; under  $\mu$  our Nh6 plays the role of that game’s Nd5; cross-type roles: our bc4 as their Pg4, our Bb7 as their pb5; transferred move h3f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#324` (ply 104) [structural distance  $d=25$ , QAP obj=22]: the master played Ne5+; under  $\mu$  our Nh6 plays the role of that game’s Nd7; cross-type roles: our bf8 as their pf5, our bc4 as their pg5; transferred move h3f2 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Nf2 (summed QAP objective 136, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



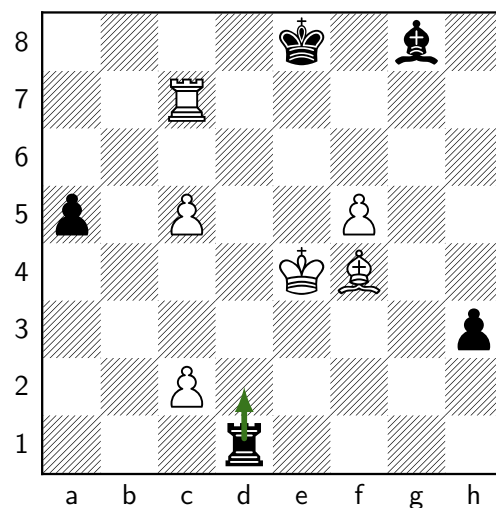
(a)



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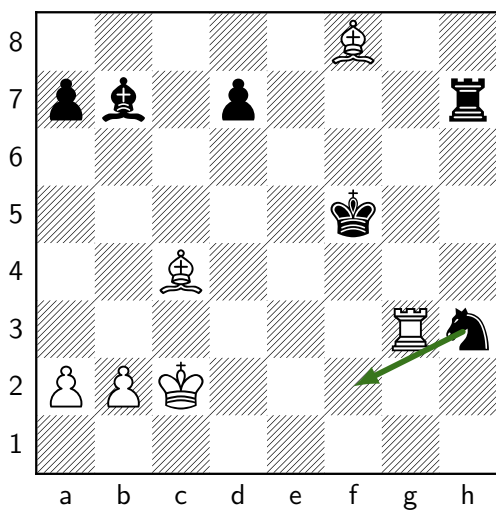


(c)

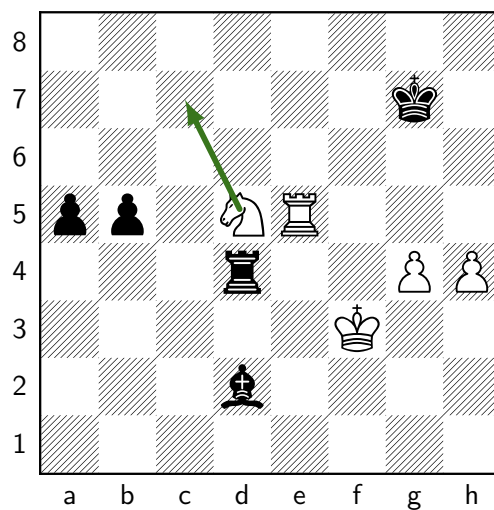


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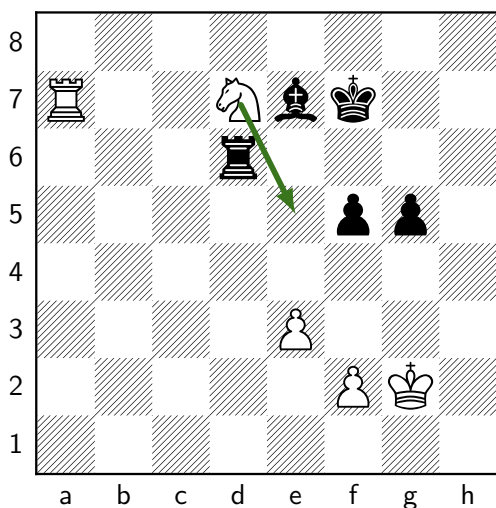
Figure 41: **Move 47, Black: Nh3**. (a) Our position; Melanie plays Nh3 (f2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1\_top\_p0000#3028 (ply 77) [structural distance  $d=21$ , QAP obj=4]: the master played Rc1; under  $\mu$  our Nf7 plays the role of that game’s Re8 (a **cross-type** correspondence); cross-type roles: our bf8 as their Pa7, our Rh7 as their bd7; transferred move f2h3 (**endorses this move**). (c) Retrieved from ccr1\_top\_p0000#1899 (ply 82) [structural distance  $d=27$ , QAP obj=3]: the master played Rg7; under  $\mu$  our Nf7 plays the role of that game’s Rh7 (a **cross-type** correspondence); cross-type roles: our Bb7 as their Pf2, our bc4 as their Pg3; transferred move f2h3 (**endorses this move**). (d) Retrieved from ccr1\_top\_p0001#977 (ply 105) [structural distance  $d=18$ , QAP obj=3]: the master played Rd2; under  $\mu$  our Nf7 plays the role of that game’s Rd8 (a **cross-type** correspondence); cross-type roles: our Rh7 as their Bg8, our Bb7 as their Pa5; transferred move f2h3 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Nh3 (summed QAP objective 37, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



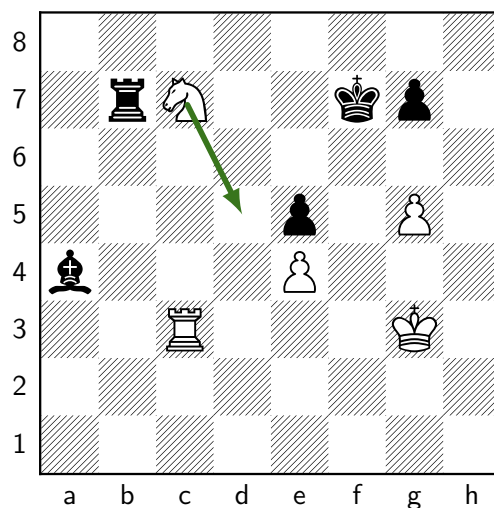
(a)



(b)



(c)



(d)

Figure 42: **Move 48, Black: Nf2 (h3f2)**. (a) Our position; Melanie plays Nf2 (h3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2459` (ply 90) [structural distance  $d=25$ , QAP obj=23]: the master played Nc7; under  $\mu$  our Nh6 plays the role of that game’s Nd5; cross-type roles: our bc4 as their Pg4, our Bb7 as their pb5; transferred move h3f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#324` (ply 104) [structural distance  $d=25$ , QAP obj=22]: the master played Ne5+; under  $\mu$  our Nh6 plays the role of that game’s Nd7; cross-type roles: our bf8 as their pf5, our bc4 as their pg5; transferred move h3f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#87` (ply 90) [structural distance  $d=23$ , QAP obj=21]: the master played Nd5; under  $\mu$  our Nh6 plays the role of that game’s Nc7; cross-type roles: our Bb7 as their Pe4, our bc4 as their pe5; transferred move h3f2 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Nf2 (summed QAP objective 108, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.